



Review article

DDoS attacks & defense mechanisms in SDN-enabled cloud: Taxonomy, review and research challenges

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ABSTRACT

Software-defined Networking (SDN) is a transformative approach for addressing the limitations of legacy networks due to decoupling of control planes from data planes. It offers increased programmability and flexibility for designing of cloud-based data centers. SDN-Enabled cloud data centers help in managing the huge traffic very effectively and efficiently. However, the security of SDN-Enabled Cloud data centers against different attacks is a key concern for cloud security professionals. Distributed Denial of Service Attacks have emerged as one of the most devastating attacks that constantly worried the entire cloud security research community. To prelude this, it is pertinent to fundamentally focus on classification of these attacks and their defence strategies in an effective way which has been the basis of this research paper. The aim of this paper is to formulate and conceptualize the taxonomies of DDoS attacks and its Défense mechanisms. Improved taxonomy of DDoS attacks highlights the various vulnerable points of vulnerability in SDN-enabled cloud architecture. Additionally, a taxonomy of defence mechanisms offers an extensive survey of recent techniques for detecting and mitigating DDoS attacks in the SDN-enabled cloud environment. Finally, we discuss the open research issues and challenges for the cloud security research community for carrying out future research and investigation.

1. Introduction

Today's computing services, including Cloud Computing, Fog Computing, Big Data Analysis, Cyber-Physical Systems (CPS) and the Internet of Things (IoT), demand higher bandwidth and speed than ever. It has been predicted that Internet users will increase to 5.3 billion by 2023 at a Compound Annual Growth Rate (CAGR) of 6% [1]. The number of Internet-connected devices will triple by 2023, with 3.6 Internet-connected devices per capita, which were 2.4 in 2018 [1,2]. Mobile users in India increased by 47 million between 2020 and 2021 [3]. IoT, connected home applications will rise with 48% CAGR, and connected car applications will grow with 30% CAGR over the period of 2018–2023 [1]. Social networks have also experienced dramatic growth in recent times, especially with the Coronavirus pandemic, as users have increased by 10%, a total of 3.96 billion by the mid of 2023 [4]. Lastly, cloud computing has also added a requirement for more flexibility and agility in networking infrastructure. Almost 67% of enterprise infrastructure will become cloud-based, and 40 zettabytes of data will be has flowed through the cloud servers and networks in 2021 [5]. Meanwhile, users can access storage and computing resources remotely, requiring a network that provides uninterrupted services. It

is inefficient with the use of the traditional network to support all these megatrends of ICT (Information and Communication Technology) [6].

Software-defined Networks (SDN) are the key solution to all the issues of current networks. The primary concept of SDN is based on the detachment of the control plane (the network's control logic) and data plane (network routers and switches) shown in Fig. 1. The data plane includes simple data forwarding devices that become active with the instructions of controller which resides in the control plane. The centralized controller is responsible for the management of the whole network; improved decision-making capability based on the global network monitoring. The key differences between traditional network and SDN are given in Table 1. The centralized networking strategy of SDN with a global view of the network is quite beneficial in the cloud computing environment. It provides the opportunity for cloud service providers to offer more flexible service to their customers by operating in multiple data centers in different locations [7]. Also, in *Virtual Switching*, vSwitches provides more visibility and control on the network, and *virtual migration* supports the migration of packets to a new virtual machine location using forwarding rules in switching.

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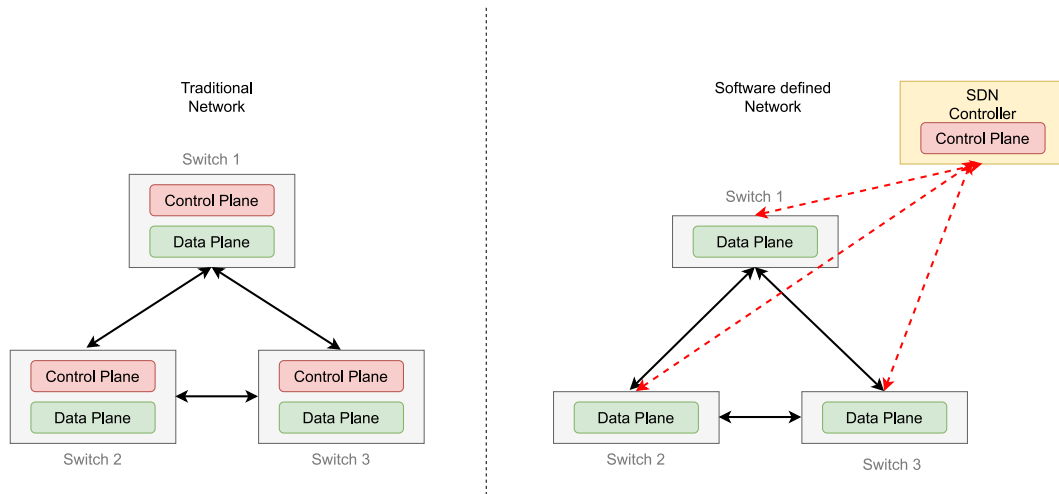


Fig. 1. Traditional Network vs. Software-defined Network.

Table 1
Comparison of traditional network and software-defined networking.

S. No.	Comparison metric	Traditional network	SDN
1.	Networking Approach	Hardware-based	Software-based
2.	Flexibility	Low — Use routers, switches and other networking equipment to create network and run the network.	High — allows to have a great command and ease of managing resources virtually using control plane
3.	Communication	Difficult — IT administrator have to arrange network services using physical infrastructure	Easy — IT administrator can direct network paths and proactively arrange network services using control plane
4.	Virtualization	Not applicable — physical location of the control plane hinders an IT administrator's ability to control the traffic flow	Applicable — Generates an abstract copy of physical network and allows to provision resources from a central location
5.	Investment in Physical infrastructure	High — more vendor specific switches are required as the size of network increases	Low — the role of hardware can be changed by the controller, i.e. same device can be act as switch and router
6.	Innovation	Low — the network is rigid	High — policy and rules deployment is fast and validation process in quite easy

Although the benefits of the adoption of SDN in a cloud environment are discussed by many researchers [8–11], the security aspect of this has not been comprehensively studied yet. SDN security has two directions — firstly, the novel features of this architecture have been used to secure the Data Center Networks (DCNs) and Cloud applications from attacks and secondly, this system is vulnerable to various cyber threats. These cyber threats includes compromised SDN controller [12], Denial of service of SDN controller [13], malicious SDN applications [14], man-in-the-middle attack in control and data plane [15], etc. Out of these threats, Distributed Denial of Service (DDoS) Attack in an SDN-enabled cloud is one, which has gained the attention of many researchers. On the one hand, SDN-enabled cloud provides protection against DDoS attacks, whereas centralized controller uses its feature of having a global view of the network to detect and mitigate DDoS attacks, on the other hand, same centralized controller needs rigid security as it is solely responsible for managing whole network of data centers and is a potential target of DDoS attack.

1.1. Motivation

SDN market is growing exponentially. According to Marketsandmarkets report, the international SDN market size will hit \$32.7 billion by the end of 2025, where it was \$13.7 billion in year 2020 [16]. The primary growth factor for the SDN market is the need of advanced

and efficient network infrastructure that can handle huge traffic and complications. Along with this, the rapid increase in demand of cloud based solutions and intent-based networking have direct and positive impact on the overall growth of SDN [17]. However, the fast adoption of SDN based network, various cyber attacks are also introduced that exploit the vulnerabilities of the network. The centralized control plane and dumb forwarding hardware in the data plane make SDN vulnerable to a Distributed Denial of Service (DDoS) Attack, which attackers can target easily. The fundamental goal of a DDoS attack is to exhaust the memory, processing, and bandwidth resources of the SDN components, which eventually results in a decline in network performance.

Distributed Denial of Service (DDoS) attacks is a hazardous threat to a network. In this, firstly, the attacker creates a connection with multiple online devices (collectively called a botnet [18]) and then bombards the target (server/website) with an excessive number of requests. As a result, the legitimate users experienced delay and in the worst case, lost the connection with the server [19]. Many organizations are migrating their services to the online platform and therefore, this attack leads to various financial losses. The attackers are committing these crimes under the influence of various motivations such as taking down the service of the competitor, ransomware, etc. Fig. 2 is showing the DDoS attack incidents that happened in the recent years. According to Arbor Networks, in 2022, there is 487 percent increase in HTTP/HTTPS DDoS attacks since 2019 [20]. Also, the strength of DDoS attacks is increasing

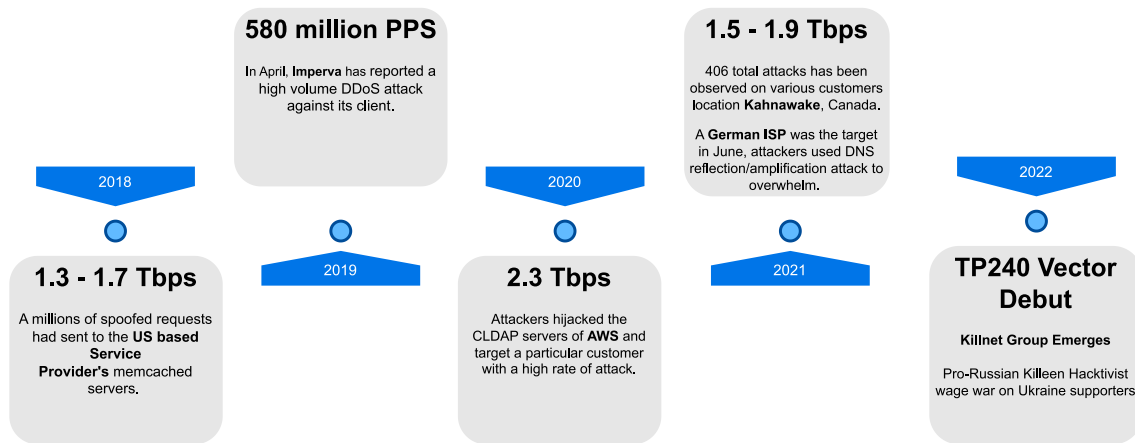


Fig. 2. Major DDoS Incidents in Recent Years.

day-by-day. In 2017, DDoS attack strength was 2.5 Tbps that surpass 3Tbps in 2021 [20].

This article is a comprehensive survey of the impacts of DDoS attacks in SDN-enabled clouds. Many of the recent research papers [21, 22] have focused on the general security of SDN and others [23–29] emphasized the DDoS attack impacts in the SDN environment only. However, in this article, we highlight the DDoS attack scenarios in SDN-enabled Cloud environment. Also, we have proposed a taxonomy of DDoS defense mechanisms in SDN-enabled cloud, which is first to the best of our knowledge. The comparison of our article with other review papers is shown in Table 2. The taxonomy of DDoS attacks in SDN-enabled cloud will enlightens the knowledge of DDoS attacks on various layers of SDN-enabled cloud including the application layer and infrastructure layer with the threat on cloud data center. Moreover, the extensive review and taxonomy of various DDoS Defense mechanisms in SDN-enabled cloud will help the researchers to understand the pros and cons of defense systems that leads to understand the properties of an ideal and robust solution to the problem.

1.2. Contributions

In this paper, we focused to deliver the detailed taxonomies of DDoS attack and defense in SDN-enabled cloud. The prime objective of the study is to deliver a comprehensive review of various DDoS defense techniques to develop a robust, effective and ideal solution to combat DDoS attacks in SDN-enabled cloud. The major contributions are mentioned below:

- **Formulating the taxonomy of DDoS Attacks in SDN-Enabled Cloud:** This taxonomy includes different types of DDoS attacks that can be launched on the SDN-Enabled cloud architecture. This taxonomy will play vital role in understanding the different vulnerable points present in different layers of SDN-Enabled cloud. This taxonomy is more detailed than the previous taxonomies given by [21,22,30].
- **Conceptualizing the Taxonomy of DDoS Defense in SDN-Enabled Cloud:** We have also conceptualized the taxonomy of different defense strategies against DDoS attacks. Various dimensions of DDoS defense mechanisms have been highlighted and discussed. This taxonomy will help the research community to develop a robust and effective detection and mitigation approach against these devastating attacks.
- **Highlighting the general research issues and challenges in SDN-Enabled Cloud:** We have discussed open research issues and challenges regarding problem. The prominent problems related to big data traffic for real-time detection, IoT-based DDoS attacks, Issues of Distributed SDN Controller Networks, and Upgradation of SDN in Internet Service Providers (ISPs) have been discussed.

- **Extensive Literature survey:** Literature discussed about the said problem is more exhaustive and elaborative that include almost 70 research papers of renowned publishers like Elsevier, Springer and others.

The present article is organized as follows: Section 2, shows the introduction to SDN and SDN-enabled cloud and its objectives. Section 3, shows the taxonomy of DDoS attacks in SDN-enabled cloud. Section 4, presents a detailed taxonomy of DDoS defense mechanisms in SDN-enabled cloud with a discussion about the research gaps of the related work. Section 5, discusses the general issues and challenges. Finally, Section 6, concludes the paper and 7 will discuss the limitation and future directions.

2. Background

It is important to understand the architecture of SDN and SDN-enabled Cloud in order to recognize the vulnerabilities in networks. SDN separates the control plane from the data plane, allowing network administrators to control network behavior through software without having to configure each individual network device manually. It typically involves a centralized controller that communicates with network devices to make real-time decisions about routing and traffic management based on the network's current state and traffic patterns. An SDN-enabled cloud refers to a cloud computing environment that incorporates SDN principles and technologies to enhance the flexibility, agility, and management of its network infrastructure. In such a cloud, SDN is used to control and configure the networking components to meet the dynamic requirements of cloud-based applications and services. This section discuss the SDN architecture and the properties of SDN-enabled Cloud in detail.

2.1. Software-defined network (SDN)

SDN reference model consists of three layers (as shown in Fig. 3), explained below:

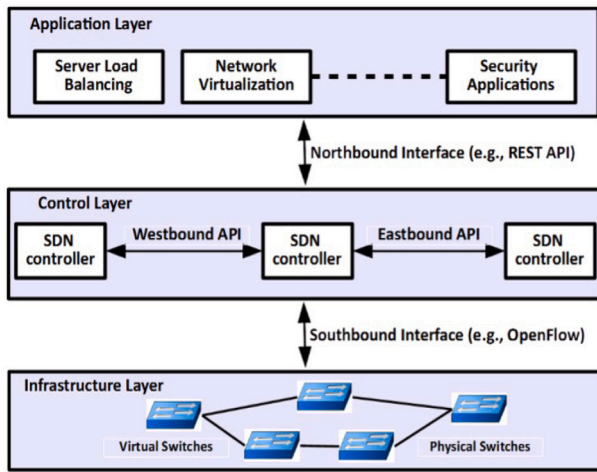
- **Infrastructure Layer:** This comprises simple networking devices which forward the incoming traffic to its destination according to the *flow rules*, an action for a particular flow, installed by the centralized controller.
- **Control Layer:** This layer contains a controller that supervises the whole network and controls the data forwarding devices using a southbound interface (e.g. OpenFlow [31]). There is a number of applications installed on the controller that provides different functionality to devices at the data plane. Also, there can be more than one controller in a network, depending upon the size of the network. These controllers communicate with each other using an east-west communication channel.

Table 2

The comparison of various review papers with our article.

Ref.	SDN SDN-enabled Cloud		DDoS attacks in SDN-enabled cloud			DDoS defenses in SDN-enabled cloud						Research Gaps	General Issues & Challenges
			Application Layer	Control Layer	Infrastructure Layer	Based on collaboration with modern technologies	Based on defense activities	Based on validation process	Based on controller scalability	Based on defense technology	Based on defense deployment		
[21]	✓	×	*	✓	✓	×	×	×	×	×	×	×	✓
[22]	✓	×	*	✓	✓	×	×	×	×	×	×	×	×
[23]	✓	×	×	×	×	×	×	×	×	×	×	×	✓
[24]	✓	×	✓	✓	✓	×	×	×	×	×	×	×	✓
[25]	✓	×	✓	✓	✓	×	✓	×	×	×	×	×	×
[26]	✓	×	✓	✓	✓	*	×	×	×	✓	×	×	×
[27]	✓	✓	✓	✓	✓	×	×	×	×	×	×	×	✓
[28]	✓	×	✓	×	✓	×	✓	×	×	✓	×	✓	✓
[30]	✓	×	✓	✓	✓	×	×	✓	✓	✓	✓	✓	✓
Our paper	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

✓— Included, ×— Not included, * — Partially included.

**Fig. 3.** SDN Reference Model.

- **Application Layer:** It includes end-user applications that enable the user to bring new functionality to the network. Examples of such applications are load balancing, security and many more.

2.2. SDN-enabled cloud

The features of SDN include bandwidth on demand, more control over the management of the network and multi-vendor operability. These properties benefit cloud computing in many ways. The amalgam of cloud and SDN improves cloud scalability and manageability. SDN-enabled cloud is a cloud in which SDN technology is used to acquire control of network infrastructure [32]. The detailed SDN-enabled cloud architecture has presented in Fig. 4. The cloud service providers include SDN in their data centers for achieving various objectives. According to [7], these aims have been classified into different categories, which are shown below:

- **Energy Efficiency:** SDN is a promising energy-saving mechanism in cloud data centers [9]. There are various ways to utilize electric energy in cloud data centers effectively. Firstly, it can be possible by switching off the server and its required cooling system when in no use because energy consumption primarily relies on the level of utilization of host servers, networking devices and cooling systems. Secondly, SDN enables all network signals to limit to a few control switches, as energy can be saved using only the essential switches rather than all in a working state. Apart from this, the combination of network and host can also be optimized to reduce electricity consumption.

- **Performance:** A network must be observed at the micro level to enhance the overall performance. There can be the alteration of routes for fast processing and reducing the traffic congestion of signals during heavy demand. SDN technique, for example, helps to boot multiple Virtual Machines (VMs), thus, reducing the deployment time of a big data center. Also, communication of minimum data, reduced computer load in physical machines and boot-up duration further speed up the deployment process. Similarly, performance can be improved by allocating dynamic bandwidth between the image library and the machine. Network performance is not just limited to the number of signals; quality and accessibility of service are also the primary factors responsible for this. SDN makes the network available whenever required. It also detects hardware issues on a network track and alters the flow differently using programmable switches and Data Center Network (DCN) connections. Moreover, SDN ensures user-targeted bandwidth, which is impossible in the ordinary network where a different type of user uses a one-line VM acceptance ratio, which means an increase in the quantity of VM in a certain cloud data library, can be enhanced by the optimum utilization of cloud and network resources [9].
- **Security:** Security is a major concern in any network system. Protecting key network components from various threats can be improved using SDN. This technology can easily analyze and respond to the overloaded traffic occurring in the system due to distributed denial of service (DDoS) Attack. SDN is effective in such situations because of its distinguished traffic analysis features, single-point control and rapid adaptability to changed network rules.

3. Taxonomy of DDoS attack mechanisms in SDN-enabled cloud

The attackers are becoming smarter and getting success in finding potentially DDoS vulnerable spots in modern network architectures including SDN and Clouds [33]. In SDN-enabled cloud, there are some weaknesses that can become significant for an attacker to launch a successful attack. Fig. 5 shows the proposed attack taxonomy and Fig. 6 is showing the DDoS vulnerable points in SDN-enabled cloud. The explanation of these attacks is given below:

3.1. Attack on application layer

The application layer of SDN-enabled cloud is responsible for handling various tasks including cloud monitoring, admission control, resource provisioning, etc. These applications are a potential target of DDoS attacks as attackers can launch HTTP floods. Also, attackers are targeting the cloud server by exploiting its resources using UDP flood, Slow HTTP test and many more.

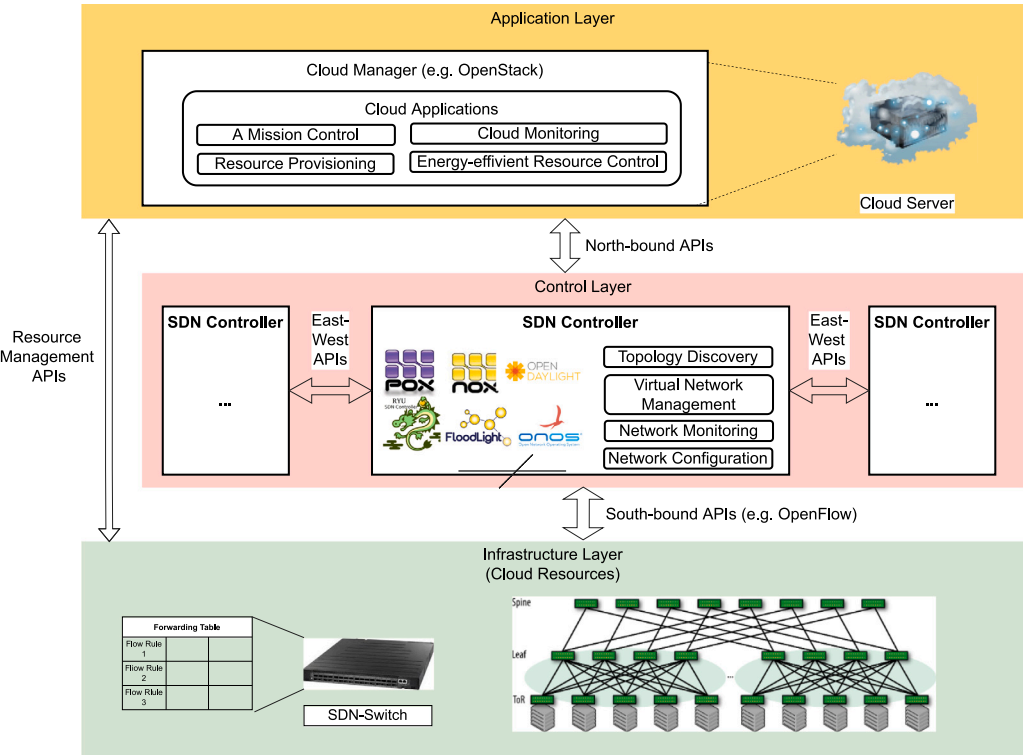


Fig. 4. SDN-Enabled Cloud Architecture.

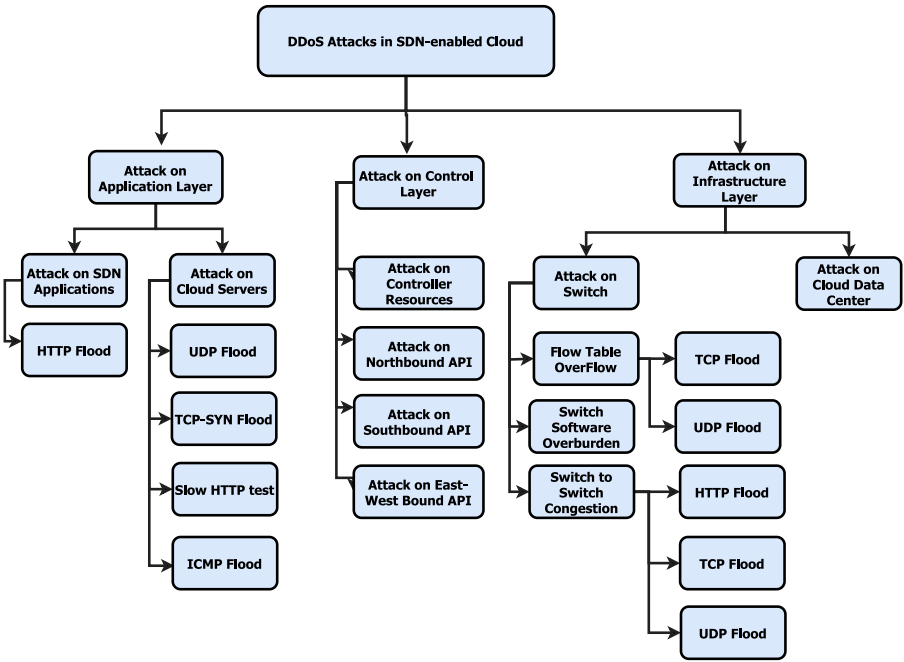


Fig. 5. A taxonomy of DDoS attack mechanisms in SDN-enabled Cloud.

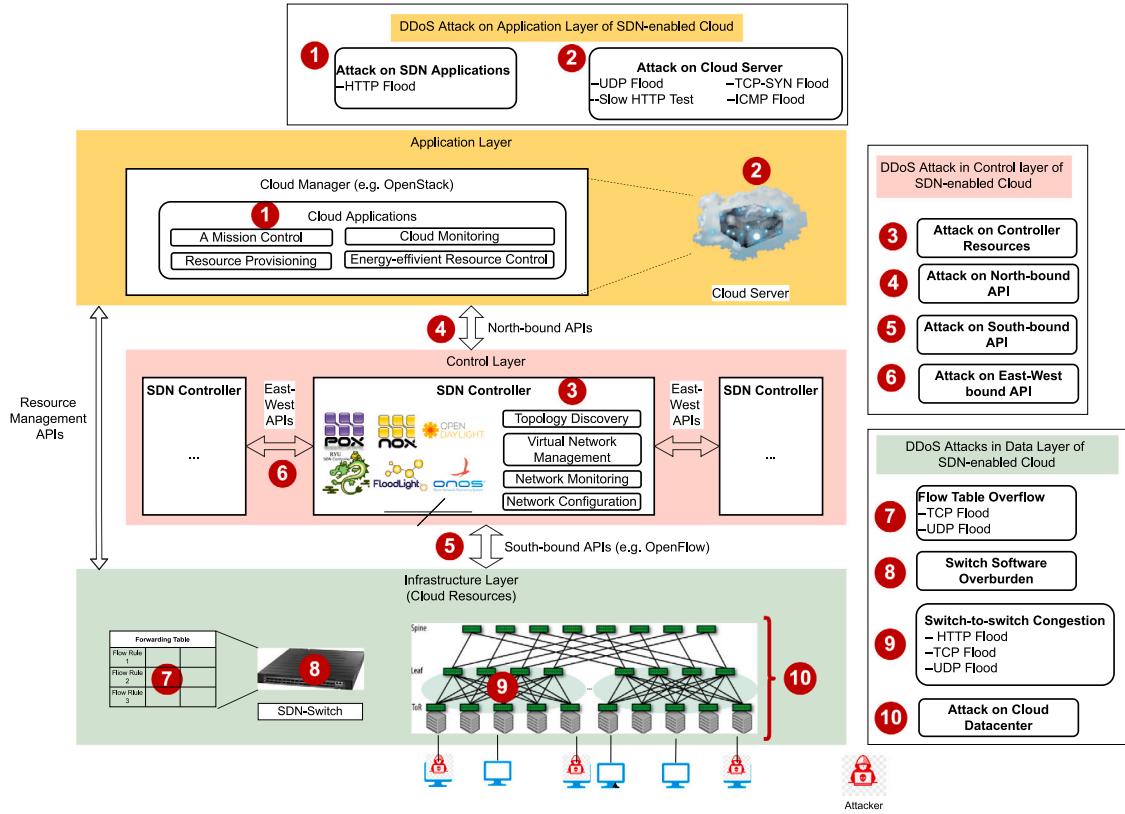


Fig. 6. DDoS vulnerable points in SDN-enabled Cloud.

3.1.1. Attack on SDN applications

An SDN application is a program that is designed to carry out some functions in an SDN environment. These applications are handled by the network administrator in order to have complete control of SDN controllers. An attack on SDN applications can hinder the normal functionality of the whole network (shown in Fig. 6). HTTP flood is a potential threat to applications of SDN.

- **HTTP Flood:** An HTTP flood attack against an application is sending a massive number of HTTP GET or POST requests [34]. These requests target the application and try to occupy as many resources as possible in direct response to each request. This task of flooding is most commonly relies on a botnet. This attack is dangerous and difficult to detect because the attack traffic is nearly indistinguishable from the legitimate traffic. In SDN-enabled cloud environment, many researchers tried to defend against this attack. Authors in [35,36] have shown how HTTP flood is targeting various cloud manager applications and hindering the normal working of the data center.

3.1.2. Attack on cloud server

Cloud server in an SDN environment is a centralized resource providing service over the Internet and gets accessed on demand by a number of users. This resource is a target of DDoS attackers and can get affected by UDP flood, TCP-SYN flood, Slow HTTP flood and ICMP flood.

- **UDP Flood:** It is an attack where a large number of User Datagram Protocol (UDP) packets are sent to the targeted cloud server that overburdens the device. The targeted server checks all UDP requests and responds to each request which results in a quick overwhelming of the server's resources, eventually a case of denial-of-service to normal traffic. Many researchers [35,37,38], have shown the attack scenarios where the cloud server in an SDN

environment is a target of UDP flood and various detection and mitigation schemes have been proposed by them.

- **TCP-SYN Flood:** In this attack, the attacker utilizes the functioning of the TCP three-way handshake protocol and consumes the resources of the targeted cloud server. In this, the attacker sends a huge number of TCP connection request that is faster than the targeted server's response, resulting in the halt of the server. The work [35,38,39], shows the attack model that targets the victim with TCP-SYN flood.
- **Slow HTTP test:** Slow HTTP attack is an exploitation of HTTP protocol, it is a protocol where a request needs to be completely received by the server before it is processed. In a slow HTTP test, the attacker sends partial HTTP requests, targets the HTTP server and aims to get a denial of service. Authors in [36] have used this attack to slow down the functionality of the web server at the application layer of the SDN-enabled Cloud.
- **ICMP Flood:** An ICMP flood attack is sending a huge number of ICMP packets against the target server that ultimately gets overwhelmed in the process of handling all incoming ICMP requests, and results in a denial-of-service condition of the server. Authors in [39,40] have opted for this attack scenario to target the cloud server in their experiment.

3.2. Attack on control layer

The control layer is a backbone of the whole network, containing controllers that are responsible for handling the network traffic and having a global view of the whole network. The attack on this layer has an adverse impact on the whole network. In this section, we have divided the attacks into the following categories.

3.2.1. Attack on controller resources

The controller is the brain of the whole SDN-enabled Cloud. When a DDoS attack, specifically flooding based DDoS attack reaches to SDN

control layer, the controller gets busy responding to the fake requests and these requests exhaust the throughput and processing capabilities of the controller. Therefore, by infecting the controller, the attacker is successful to degrade the whole network of SDN-enabled cloud.

3.2.2. Attack on northbound API

A northbound API is a connection between the application layer and the control layer. This API supports a wide variety of applications. This link is a target for attackers in case if they want to target the application layer of the SDN-enabled cloud.

3.2.3. Attack on southbound API

A southbound API is a connection between the control layer and the data layer. It is used to have communication between the controller and the switches and router of the network. This link is always required to be in a working state in order to have un-delayed communication between the controller and switches. When the *flow table overflow* (mentioned in 3.3.1) condition is there, a switch starts sending the whole packet toward the controller. This results in higher bandwidth consumption and the normal flow requests experience delay due to congestion.

3.2.4. Attack on east-west bound API

This link allows communication in multiple controllers at the control layer. This channel is quite crucial in order to have a stable and efficient working of SDN-enabled cloud network having multiple controllers. The attack on this communication channel initiates when an attacker targets multiple controllers. This attack not only consumes their resources but also congests the connection between them.

3.3. Attack on infrastructure layer

This layer of SDN-enabled cloud includes forwarding devices that receive instructions from the centralized controller and forward the packets according to the flow rules installed by the controller. There can be multiple attacks on this layer such as an attack on a switch, where flow table overflow is one of the major threats and an attack on the cloud data center.

3.3.1. Attack on switch

In a traditional network, a switch works independently of the rest of the network and it has its own brain to take decisions. But, in an SDN environment, a switch is simply a hardware device which works according to the instructions given by the SDN controller. This not-so-intelligent device has some vulnerabilities that are exploited by attackers such as flow table overflow and switch software overburden.

- **Flow Table Overflow:** The limited storage of switch is able to store a limited number of flow rules (up to 2000 flow rules). So, the attackers take advantage of this and send attack traffic towards the switch which ends up storing many flow rules. In the worst case, the table becomes full and leads to the dropping of packets. TCP and UDP floods have been used by attackers to implement this attack.
- **Switch Software Overburden:** The OFA (OpenFlow Agent [41]) in the OpenFlow switch is responsible for generating flow requests for the table miss entries in the flow table. When DDoS attackers send multiple malicious requests at the same time, then this OFA gets overburdened because of its minimal capacity of generating requests.
- **Switch to Switch Congestion:** Also, when there are multiple targets under one to more switches, they can attempt to congest the communication channel between switches too. TCP, UDP and HTTP floods have been used by attackers to implement this attack.

3.3.2. Attack on cloud data center

Data Center downtime due to DDoS attack is very common these days. The unavailability of the data center is becoming a crucial concern as this not only leads to incurring high costs and time but also, damages the brand reputation and customer trust. A high volumetric DDoS attack is able to saturate the links of even the largest data center. One example is the Mirai Botnet [32], where the attack size was in Terabytes.

4. Taxonomy of DDoS defense mechanisms in SDN-enabled cloud

DDoS attacks are a potential threat to SDN and SDN-enabled clouds. It is becoming quite challenging to differentiate the normal packets from malicious/spoofed packets that are coming from distributed sources. Although SDN has the capability to defend DDoS attacks on a large scale, the centralized architecture itself is a DDoS target. Therefore, first, there is a need to save the architecture from this attack. Researchers are in a continuous effort to detect and mitigate this attack by opting for various defense techniques. Some of these are the combination of new technologies such as big data and blockchain, others, are based on Machine and Deep learning or information theory and so on. We have proposed a novel DDoS Defense taxonomy in the SDN-enabled cloud, shown in Fig. 7. To our best knowledge, this is the first attempt to analyze the various aspects of DDoS attack, particularly in SDN-enabled cloud in a comprehensive manner. The discussion and relation of each section for defense are mentioned below:

- The present technologies in defense of DDoS attacks are playing a crucial role. *Modern technologies* including Fog/Edge Computing [42,43], Big Data [44,45], Blockchain [46,47], etc. have been explored by scientists to effectively reduce the devastating impacts of DDoS attacks. This section not only mentions DDoS detection and mitigation in the case of conventional networks but also portrays the importance of defense in SDN-based networks and cloud environments.
- Apart from this, another section discusses the techniques based on *strategy selection*. There are authors who are exploring the various strategies based on Artificial Intelligence (AI), Statistical methods and Information theory. The AI-based Machine learning and Deep Learning techniques are thoroughly explained with various defense schemes, attack types like flooding-based or low-rate, Performance metrics and a summary of related works. Also, the reason for the supreme adoption of particular defense techniques is mentioned in the discussion part. Similarly, a detailed discussion on statistical and information theory-based defense mechanisms has been discussed.
- Next, researchers developed their defense technique based on *scope*. For example, some emphasized the significance of DDoS Detection [35,38,48], while others importance of decreasing the attack impact, i.e. DDoS Mitigation [49]. To investigate the detection rationale is a primary step towards the initiation of DDoS detection. It is a signal or indication that differentiates the attack and legitimate traffic. In this section, the detection rationales mentioned by various authors are discussed separately with the description of DDoS detection scheme.
- Moreover, one section discussed the worth of *validation processes* of proposed defense techniques which consist of simulation/emulation, real testbed and mathematical models. It is vital to understand the reason for the adoption of simulation and emulation-based software for experiments and results. It has been mentioned in the section that the majority of research is based on Mininet [50] and NS3 [51] rather than a real environment. This raised the concern of having accurate results for various defense techniques.

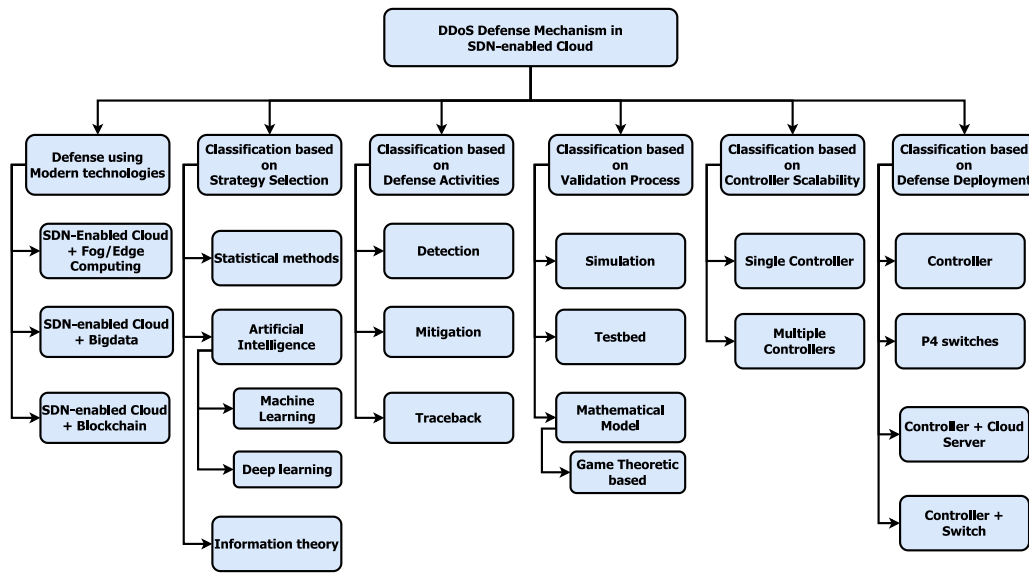


Fig. 7. A taxonomy of DDoS Defense Mechanisms in SDN-enabled Cloud.

- Along with this, defense based on *controller scalability* has also been discussed. Single-controller architecture topology is opted by the majority of researchers to conduct the experiments instead of multi-controller architecture. It has been discussed that current large networks (e.g. ISP networks) work on distributed controllers. So, the lack of research on the security of these multi-controller networks is a major research issue. A relevant discussion of each sub-section along with pros and cons is given with a related literature review.
- Last, researchers selected various *defense deployment spots* where their respective defense mechanisms will get activated and perform DDoS detection and mitigation. Some have selected a controller, other P4 switches, also the combination, such as Controller and cloud server, and Controller and Switch. Each deployment has advantages and disadvantages such as if the defense technique is deployed on the controller then the resources and memory of the controller are consumed at a high pace as it is responsible for the task of network supervision as well as detection of DDoS attacks. Other issues related to the dependency of individual switches and the complexity of the collaboration of controller and switch have also been discussed.

4.1. Defense using modern technologies

The technologies including big data, blockchain and edge computing are involved in the successful transformation of traditional industry into the current industry, i.e. Industry 4.0. These trends are also helpful in providing security to clouds, especially against DDoS attacks in SDN-enabled clouds. Many authors have utilized the advantages of these megatrends and proposed collaborated defense techniques. Accordingly, we have divided these into SDN-enabled Cloud + Fog/Edge Computing, SDN-enabled Cloud + Big Data and SDN-enabled Cloud + Blockchain. Table 3. is showing the summary of these research works.

4.1.1. SDN-enabled cloud + fog/edge computing

Fog and edge computing is providing benefits to the cloud computing by providing some features including decentralization and capability of local analysis of traffic with minimal resources. Along with these advantages, this paradigm can defend DDoS attacks in SDN-enabled cloud. Priyadarshini et al. [42] have shown that with the collaboration of Fog and SDN, the detection and mitigation of DDoS attacks is quite effective in a cloud environment. The authors have used

a floodlight controller [52] which is hosted on the fog server in the fog layer. Whenever a request comes, the controller classifies the traffic as genuine or malicious based on the trained dataset of three machine learning algorithms. To validate their work, they have used three physical machines for each layer (Cloud, Fog and Application). They have found that the KNN algorithm outperforms with 86% accuracy. Sadiq et al. [43] show that a spoofed-based technique is a primary way to implement DDoS attacks and the authors address this particular malicious technique in their work. They have shown how cloud and fog are vulnerable to security challenges and SDN is a promising way to mitigate DDoS attacks. They have proposed a theoretical framework where they use the SDN concept on fog servers where SDN controllers use classifier to detect maliciously and attack traffic.

4.1.2. SDN-enabled cloud + big data

The data processing using traditional methods in modern networks is ineffective, especially to process a large amount of data [44]. Also, big data-based machine and deep learning techniques require big data-based solution to process data and make predictions in real-time with high accuracy, robustness and efficiency. A big data-based DDoS detection scheme has been proposed by Yungaicela-Naula et al. [44]. They have addressed the limitations of threshold-based and feature classification-based detection schemes. In their proposed work, they have Apache Kafka to stream traffic statistics in real-time, and apache Spark to pre-process the data and select a specific ML-based model from Machine Learning (ML) pipeline. The evaluation is done on a real testbed. Apache Spark is implemented by [45] to detect and mitigate DDoS attacks in large amounts of traffic. The authors mentioned Tshark as a capture server that captures the traffic and produces a traffic log. Then, Spark Cluster detect the attack traffic based on log files. The router application mitigates the DDoS attack traffic by blocking it.

4.1.3. SDN-enabled cloud + blockchain

A blockchain is a distributed ledger that can record transactions in an efficient and permanent way. Blockchain has been implemented by many researchers to build secure systems and applications. Recently, researchers have used blockchain to handle DDoS attacks in SDN-enabled cloud and IoT environments [53]. Rehman et al. [46] added blockchain technology to SDN-enabled cloud to make the environment secure. They have compared both OpenFlow based SDN-enabled cloud and blockchain-based SDN-enabled cloud in terms of throughput, CPU utilization and response time. They have used Mininet for simulations

Table 3

The summary of the articles having defense using modern technologies.

Ref.	Vulnerable spots	Defense scheme	Notable points
[42]	Cloud using traditional network	SDN + Fog based Detection	A fog based detection scheme is used where a fog defender module is trained by a machine learning algorithm.
[44]	SDN enabled cloud	SDN + Big Data	Evaluation parameters are related to resource utilization, i.e. CPU, Memory and Disk. The apache kafka performance decreased as the record size increases.
[45]	Cloud using traditional network	SDN + Big data	The simulation results shows that big data is extremely useful to detect DDoS attacks in high volume traffic data. The combination of SDN and big data is proved to be a highly efficient automatic DDoS defense system.
[46]	SDN enabled IoT cloud	SDN + Blockchain	During DDoS attack, the proposed technique is able to protect in less time with minimal utilization of CPU. The response time comparison of Openflow and SDN + blockchain technique has been carried out, where the latter is giving better performance.
[47]	SDN based IoT network	SDN + Blockchain	Blockchain-based solution as a load-balancer to service millions of IoT devices. This technology has been used to represents rights to access resources and to transfer rights from one user to another.
[54]	SDN enabled cloud	SDN + Blockchain	sFlow has been used to collect the statistics of traffic. Ethereum blockchain is a decentralized platform for developing smart contracts Detection rate is: 100%, False Positive Rate (FPR): 26%.
[43]	Cloud using traditional network	SDN + Fog based Detection	Floodlight Controller has been used to detect DDoS attack. The KNN algorithm is giving highest accuracy.
[55]	SDN enabled cloud	Cloud + Edge Computing	The collaboration of cloud and edge for DDoS detection has reduced the burden on SDN controller. Cloud has been used for parameter calculation. Edge controller focus on traffic detection to reduce the workload.
[56]	SDN Layers	SDN + Quantum Cryptography	The simulation result shows the accuracy of 92%. Secure the TLS protocol for a reliable and efficient communication between SDN switches and Controller.

and Wireshark to analyze the traffic. Houda et al. [54] have proposed a mitigation technique based on blockchain. They have provided DDoS defence within the SDN domain and in communication with other domains. For mitigation at decentralized SDN domains, they have used the concept of smart contracts to efficiently facilitate mitigation with low cost. Faizullah et al. [47] have proposed a secure SDN environment for IoT devices using permissioned blockchain. It is impossible to corrupt the blockchain due to the replication of the ledger. They have shown how a blockchain technology is providing decentralized trust. The results are showing that it outperforms the other baseline models.

4.2. Classification based on strategy selection

Researchers are using various techniques and technologies to develop an efficient and effective defence mechanism against DDoS attacks. The various defense strategies based on machine learning, deep learning and entropy have been used by a number of researchers. A detailed description of their proposed detection and mitigation techniques is given below.

4.2.1. Statistical methods

Various statistical techniques are used by researchers to detect and mitigate DDoS attacks in SDN-enabled clouds. Batool et al. [57] used a weighted moving average and standard deviation approach to minimize the false-positive rate of TCP-SYN attacks. They have used a multimodular detection approach where they have counted the SYN requests, and when these requests become higher than the threshold. Shohani et al. [58] mentioned four-stage detection techniques. First, extensive experiments have been done to observe attack and normal traffic, and then a statistical trapezoid model estimates the count of table misses. After that, linear regression estimates the threshold of the table misses. Finally, the detection of attack is carried out by the

derived model. The validation results show that the method can detect DDoS attacks at its early stage.

4.2.2. Artificial intelligence (AI)

Artificial Intelligence is a simulation of human intelligence processes by machines, especially computer systems. These futuristic techniques are quite effective to predict and detect DDoS attacks. These include machine learning and deep learning.

- **Machine learning (ML)** solves complex problems in many fields. The detection and mitigation techniques based on machine learning is better than statistical-based and signature-based techniques. The ML-based classification techniques are trained to detect malicious traffic with high accuracy [64]. The summary of these techniques is shown in Table 4.

Random Forest (RF) is an ensemble classifier with low classification error that consists of many individual decision trees uncorrelated, which operate as a single set with the power to handle large datasets and noise in the data. Naula et al. [36] have compared this algorithm with various machine learning and deep learning algorithms to detect DDoS attacks in SDN-enabled cloud environments. The results show that RF is showing low accuracy compared to other techniques. The experimentation is done on the ONOS controller. Dehkordi et al. [59] implemented this algorithm to classify the attack and normal traffic. After entropy computation, the training samples with extracted features have been given as input to this algorithm. *Support Vector Machine (SVM)* is a supervised learning model that can perform non-linear classification. This algorithm can detect intrusions with limited training data. Naula et al. [36] have compared this algorithm with machine learning and deep learning algorithms to detect DDoS attacks in SDN-enabled cloud environments. The results show

Table 4

The summary of various works related to Machine learning.

Ref.	Defense scheme	Attack type	Performance metrics	Notable points
[36]	KNN, SVM, RF	HTTP Get Flooding, TCP SYN Flood, UDP flood, DrDNS attack	Detection Accuracy — (KNN: 99.97%, SVM: 99.77%, RF: 96.36%), Precision: (KNN: 99.97%, SVM: 99.77%, RF: 98.51%), Recall: KNN: 99.97%, SVM: 99.77%, RF: 96.36%, FPR: KNN: 1.816%, SVM: 10.23%, RF: 7.832%)	Able to detect transport and application layer DDoS attack. D-ITG tool has been used to generate benign traffic. The detection accuracy gets decreased when the network topology gets changed.
[38]	Fuzzy Hybrid with AI	TCP and UDP Flood	Detection Accuracy — 95%	Accurate detection for high traffic scenarios. The defense scheme might increase burden on controller.
[45]	Discrete scalable memory based Support Vector Machine (DSM-SVM)	—	Detection Accuracy — 99.7%	It provides protection to Control-data Communication link.
[54]	Bayes theorem	UDP and ICMP flood attack	Accuracy — 100%, FPR — 26%	sFlow has been used to collect the statistics of traffic.
[59]	BayesNet, J48, Random Tree, Logistic Regression, REPTree	—	Accuracy — 99.85%, FPR — 0.1%	It can detect low volume attacks also. Used only one controller.
[60]	SVM, c4.5 Decision Tree and Naïve Bayes	Low — rate DDoS attacks	Accuracy — 93%	Detect TCAM Saturation Attack. Only detects low rate attacks.
[61]	AdaBoost Boosting Algorithm	—	Detection Accuracy — 99.2%, FPR — 0.9%	Decision Stump is used to generate the learning data and serves huge data to the servers within less time. The proposed technique can detect attack in low communication also.
[62]	Probabilistic Graphical Model	ICMP and HTTP flood	Accuracy — 89.30%, FPR — 10.70%	Opted a model update approach, where the inference model is updated on periodic basis. Used real traces to evaluate the performance of the mitigation module.
[63]	SVM	HTTP Flood	Accuracy — 99.8%	Sniffer has been used a collector for statistical information. Entropy has been used a initial phase of detection.

that SVM is showing low accuracy compared to other techniques. The experimentation is done on the ONOS controller. A machine learning-based detection scheme is utilized by Revathi et al. [45]. The authors protected the controller from control-data communication congestion using SVM classifier. A semantic multilinear component analysis algorithm is applied for feature extraction, and the validations on the KDD dataset show 99.7% detection accuracy. A comparison between various machine learning detection algorithms is done by Sudar and Deepalakshmi [60]. The research focuses on the detection of low-rate DDoS and TCAM saturation attack. A comparison of SVM and other algorithms is made, and the results show the high accuracy of SVM. Lingfeng and Hui [65] include this technique to make a DDoS recognition model and claim the scheme has high robustness as it solves the optimization problem. In the initial phase, entropy has been calculated, and the traffic statistics have been sent to the SVM classifier to identify the attack traffic. The results show high detection accuracy as compared to the previous works.

K-Nearest Neighbor (kNN) determines the new data class by looking at its nearest K neighbors. KNN has shown high performance in detecting attacks. Naula et al. [36] have compared this algorithm with various machine learning and deep learning algorithms to detect DDoS attacks in SDN-cloud environments. The results show that SVM is showing low accuracy compared to

other techniques on ONOS. Priyadarshni et al. [42] have created a DDoS defender module on the SDN controller. This module has compared many machine learning algorithms, where kNN outperforms other classifiers with an accuracy rate of 87.343%. Researchers have used many *Decision Tree (DT)* variants to detect DDoS attacks efficiently. C4.5 is one of them, which classify by constructing trees using if-then rules. Suadar and Deepalakshmi [60] have compared this algorithm with various machine learning algorithms to classify the attack and normal traffic. C4.5 Decision Tree can gain the highest accuracy, i.e. 91%. The proposed work can also detect low-rate attacks very efficiently. A *Naïve Bayes classifier* works on the probability of each feature. Sudar and Deepalakshmi [60] mentioned the classifier to detect low-rate DDoS attacks, where it is compared with SVM and Decision Tree. This algorithm stood second in terms of accuracy, where SVM outperforms others. Also, [54] implemented Bayes theorem to classify traffic as legitimate or illegitimate. The algorithm is first trained and then classifies the traffic. The detection rate is 100%, and the FPR is 26%.

Fuzzy logic is an approach to computing based on “degrees of truth” rather than the usual “true or false” (1 or 0) Boolean logic on which the modern computer is based. A fuzzy Inference System and Artificial Neural Network is implemented to classify attacks and normal traffic [38]. The work mentioned fuzzy rules to find

malicious packets — low successful packets, high packet loss and high error rate. The adaptive Boosting Algorithm is implemented as an ensemble method. Tufa et al. [61] is implemented to coordinate with SDN features to detect DDoS attacks. The experimental results show high accuracy, also when there is low communication. The *graphical Model* is a unifying framework for describing the statistical relationships between large collections of random variables. XIAO et al. [62] mentioned a feature-pattern graph model to detect DDoS attacks in an SDN environment. The work employed network patterns as nodes and similarity as weighted links. Metric Learning and Mahalanobis distance are applied to model the similarity between nodes. The technique can find unknown attacks and is scalable for inserting new nodes into the graph model via local and global updates.

- **Deep learning (DL):** Deep learning-based models are built on a series of nonlinear processing layers to learn many levels of data representation from a large volume of labeled samples [35]. Therefore, these techniques are highly efficient in detecting DDoS attacks. The summary of these works is shown in Table 5.

Extreme Learning Machine (ELM)-based model is a training algorithm for single hidden layer feedforward neural network (SLFN), which converges much faster than traditional methods and yields promising performance. Ravi and Shalinie [64] have proposed a scheme that is able to detect DDoS attacks in an IoT environment. They have mentioned some limitations of single-layer ELM, therefore, they have used a Semi-supervised deep extreme learning machine (SDELM) model for detection. Multilayer Perceptrons (MLP) is a supplement to the feed-forward neural network. It consists of three types of layers—the input layer, output layer and hidden layer. Naula et al. [36] have compared this algorithm with various machine learning and deep learning algorithms to detect DDoS attacks in SDN-cloud environment. The results show that MLP is one of the techniques having high accuracy. They have used an ONOS controller for experimentations.

Self-Organizing Maps (SOM) are used for clustering and mapping (or dimensionality reduction) techniques to map multidimensional data onto lower-dimensional which allows people to reduce complex problems for easy interpretation. [37,55]. A prevention technique is proposed by Harikrishna and Amuthan [66] based on Self-organizing maps. The scheme can learn constantly and improve the adaptation to classify the normal and attack traffic. The technique is validated on a testbed and the traffic rate is between 100 to 500 Mbps. Harikrishna and Amuthan [37] have proposed a Convolutional Recursively Enhanced Self organizing map mitigation technique that dynamically changes the rules and data investigation for high reliability. Also, the scheme successfully provides a high learning rate and initialization process for better categorization of traffic flows. A deep learning-based Spatial-temporal graph convolutional network has been used by Cao et al. [35] to detect DDoS attacks in data center network. The work considered the in-bound Telemetry to collect the network state parameters. Then, the spatial and temporal features of the data have been found, and the detection model finds the switches under a DDoS attack.

4.2.3. Information theory

Entropy belongs to the degree of randomness, which means the higher the randomness, the more is the value of entropy and vice versa [57]. Researchers have used entropy for DDoS defense and a summary of the related work is given in Table 6.

Abdulqadder et al. [38] have used entropy at the controller to detect DDoS attacks in SDN based 5G networks. The calculated entropy of the destination IP is carried out because a DDoS attack occurs on a specific destination IP address. Therefore, entropy can detect DDoS when the entropy of the destination IP address decreases and the

source IP address increases. Mishra et al. [48] used a threshold-based detection scheme to mitigate flooding-based DDoS attacks on the POX controller. The work mentioned three thresholds, one for incoming traffic, a second for entropy threshold and a third to mitigate the attack flow. An entropy-based detection and mitigation technique has been proposed by Nadim and Foysal [68]. The authors employed hash tables to make the entropy calculation easier and more effective. A POX controller has been used to check the proposed scheme's performance. Dehkordi et al. [59] have used different datasets to evaluate their entropy-based detection technique. Two thresholds have been appointed, static and dynamic, to detect the attacks. The proposed technique is validated using each dataset, and the average detection accuracy is high compared to the previous works. Mishra et al. [48] have proposed a defensive technique for DDoS attacks based on entropy variation. They mentioned the counter parameter of the flow table to recognize DDoS attacks. A counter value is the number of hits for a particular flow, higher in a DDoS attack than a non-DDoS attack. First, Guesmi and Saidane [69] used adaptive fast entropy to detect DDoS attacks in cloud servers. Along with this, the work shows the inbuilt capabilities of SDN to defend DDoS attacks. The scheme can predict the threshold and announce the attack using various fields of the global table of controllers. Tsai et al. [70] proposed entropy-based detection to detect DDoS attacks in the SDN-enabled cloud. An SDN application has been installed on the Ryu controller, which monitors with consecutive intervals. Also, NIDS (Network Intrusion Detection System) forward the attack flow, so that the model can detect the attack prior and stop it.

4.2.4. Other

Also, researchers have used techniques other than the above-mentioned approaches. A *sinkhole strategy* has been used by the authors to mitigate TCP-SYN attacks. According to this mitigation strategy, one port has been declared as a sinkhole and, the attack flow has been given the lowest priority and forwarded it to the sinkhole port [57]. A hybrid technique based on *Moving Target Defense (MTD)* and *Deception scheme* has been proposed by Zhou and Cheng [39]. This technique leads the attacker to take the wrong decision and consume their resources. Then, an optimal defense selection strategy selection algorithm has been proposed to optimize the defense implementation. This lightweight scheme is been validated using simulation. The proposed scheme leads to the CPU utilization of SDN Controller till 65% during the attack and, performance and overhead for real services is less than 20%. The authors have focused on external DDoS attacks only. Bhushan and Gupta [49] have used the flow tables of other switches to handle the attack traffic. By this, the burden of the switch under attack becomes less and the holding time of SDN gets increases. This approach is proposed to defend against flow-table overloading attacks. They are providing the solution with minimal involvement of the controller.

Discussion: Out of Artificial intelligence, Entropy and Statistical models, the majority of researchers have developed artificial intelligence-based defense systems. As shown in Fig. 8, machine learning opted for maximum time with 41.5% and among all classifiers, SVM is the most used one. The reason behind this is it maximizes the likelihood of linear separation and also it is quite effective in high-dimensional spaces. Therefore, it is better than all linear classifiers. Apart from this, the majority of detection schemes are based on setting a threshold, which means several traffic indicators such as traffic rate, maximum entropy, and packet delay are kept track of; if the indicators exceed a predetermined threshold, then an attack may occur in the network. This approach results in high false-error rates because of a hard threshold [44]. Therefore, there is a requirement for more robust and efficient techniques that can reduce this collateral damage. Also, the majority of the research work is carried out to provide security to the control layer only. Only a few have highlighted the importance of providing security to the infrastructure layer. It is easy to target switches because these have low computation capability and low memory therefore, it is easy to overload them in minimum time by flooding

Table 5

The summary of various works related to Deep learning.

Ref.	Defense scheme	Attack type	Performance metrics	Notable points
[35]	ST-GCN	TCP flood, UDP flood and HTTP Flood	Detection Accuracy: (TCP flood — 90.5%, UDP flood — 89.1%, HTTP flood — 90.5%), FPR-(TCP flood-0.050, UDP Flood — 0.057, HTTP flood — 0.041)	Protects data plane from flow table overflow attack. In mitigation, all switches coordinates in attack mitigation, therefore, reduce the burden one edge switch.
[36]	MLP, CNN, GRU, LSTM	HTTP Get Flooding, TCP SYN Flood, UDP flood, DrDNS attack	Detection Accuracy: (MLP: 99.89%, CNN: 99.13%, GRU: 99.80%, LSTM: 99.87%), Precision: (MLP: 99.89%, CNN: 99.33%, GRU: 99.87%, LSTM: 99.87%), Recall: (MLP: 99.89%, CNN: 99.13%, GRU: 99.88%, LSTM: 99.87%, FPR: (MLP: 1.503%, CNN: 4.392%, GRU: 4.454%, LSTM: 1.571%))	Able to detect transport and application layer DDoS attack. D-ITG tool has been used to generate benign traffic. The detection accuracy gets decreased when the network topology gets changed.
[64]	ELM based semi-supervised Machine Learning	UDP flood with spoofed IP	Detection Accuracy — 96.28%, Average Detection time: 2.3 ms, Switch Controller Bandwidth Consumption — 30%, Detection Time of Malicious IoT — 2.7 s	The model is trained with both labeled and unlabeled dataset. This trained model is deployed on all local controllers.
[66]	CRESOM	UCP, TCP and ICMP attack	Classification Accuracy — avg. of 21%, FPR: reduced by 19%	The scheme is able to do constant learning and improve the adaptation to classify the normal and attack traffic.
[67]	SOM	UDP, TCP and ICMP flooding DNS and IP Spoofing	Detection Accuracy: 99.88%, TPR, FPR and TNR is minimized on an average of 11.25%.	Able to detect the DDoS attack in early stage. The traffic rate is between 100 to 500 mbps.

Table 6

The summary of various works related to Information Theory.

Ref.	Entropy Type	Detection Metric	Attack type	Summary
[38]	Entropy	Destination IP	TCP and UDP Flood	Using entropy they are able to detect DDoS attack in minimal time and after that, the traffic has been forwarded to the classification module.
[39]	Hash table based entropy calculation	Based on destination IP	UDP flood	Hash table has been used to make the entropy calculation easy for each incoming packet of all the switches. They have used only destination IP as a detection metric.
[48]	Entropy	Based on destination IP, protocol, destination port number	–	Calculated entropy for protocol, destination IP and Source IP and then processed the packets according to the order of entropy with descending order.
[54]	Shannon Entropy	Destination IP, Destination Port Address, Source IP address	TCP and UDP flood	sFlow protocol is used to collect the traffic information and entropy has been imposed to compute entropy.
[57]	Entropy	Destination IP	TCP-SYN Flood	The Window size is 50 and threshold value of entropy is 1. The entropy rate has increased to 700 per second after the mitigation process while it is 300 under attack flow.
[69]	Adaptive Fast Entropy	Flow Count	ICMP and TCP flood	The proposed scheme is able to detect and block the attack traffic within 15 s, when the count of packets reaches 25 000/s. The comparison has been done with various schemes and the proposed outperformed them.
[70]	Entropy	Destination IP	ICMP flood	A dynamic way has been opted, where the application dynamically changed its threshold with a classified algorithm. For both external and internal attacks, the window size is 50 packets.

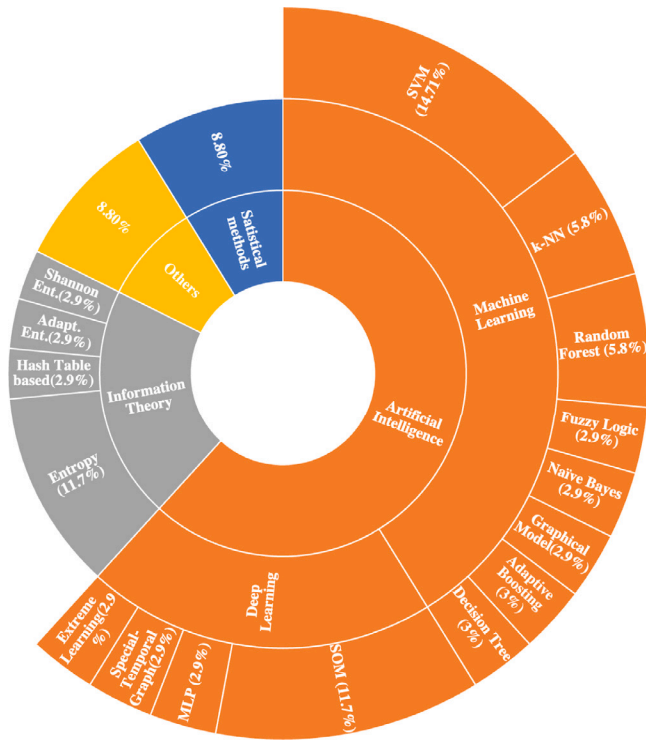


Fig. 8. Classification of defense techniques based on strategy.

them with high attack traffic. Moreover, minimal work has been done to save various APIs including east-west bound and northbound along with SDN applications. All these domains of SDN-enabled cloud are crucial to be in a working state for the normal functioning of the network. Therefore, providing defense solutions to these parts of the SDN network is also a major research issue. Most of the detection schemes are able to outperform flooding-based DDoS attacks only. Only a few are working to detect and mitigate low-rate DDoS attacks [71]. Most of the detection schemes are based on pattern matching, flow rate and flow count, therefore, low-rate attacks remain undetected. Therefore, proposing a defense against low-rate DDoS attacks is also a concern for the researchers.

4.3. Classification based on defense activities

It is important to detect DDoS attacks and react against it for the reduction of its impact on the network. In this section, the categorization is on the basis of defense activities, i.e., Detection, Mitigation and Traceback.

4.3.1. Detection

The detection of a DDoS attack is about sensing the occurrence of the attack. This phase is quite important in order to take further action. In this process, detection rationale plays vital role. It is defined in many means, such as a threshold for packet count or packet loss rate [38], a feature selection based on data [35], entropy calculation based on destination IP [48] and so on. Detection Rationale is a reason which researchers differentiate between attack and legitimate traffic. Some of the researchers have clearly mentioned this in their proposed work and these are listed in Table 7.

4.3.2. Mitigation

The mitigation of DDoS attack is to eradicate the effects of the attack. It is the final step for many of the attack defenses and therefore, it is showing the effectiveness and performance of defense techniques.

The crucial step in mitigating DDoS attacks is to minimize collateral damage, i.e., while filtering DDoS attack traffic, the legitimate traffic should not suffer. Researchers have used many techniques to reduce the impact of attack including inserting a flow rule of dropping the attack traffic, placing honeypots, Insertion of a drop rule, Traffic rate limiting, Using Blockchain, etc. [54] have used blockchain to mitigate the DDoS attack traffic in an inter-domain environment. It utilizes the concept of smart contracts to make coordination with other SDN Autonomous Systems.

4.3.3. Traceback

The traceback of the attack is about finding the attack source location. A few efforts have been done by the researchers to trace the attack sources. Aggarwal and Tapaswi [71] are finding the attack source by using the record-route option in IP protocol, based on the DPM scheme. The scheme is about enabling the record-route option for every flow and the router stores one of its IP addresses. If the field is not set, it is declared malicious. The target server analyses the IP addresses stored in the option header field and finds out the attack source using a virtual network maintained by the SDN controller.

4.4. Classification based on validation process

To check the performance of the proposed detection and mitigation technique in network security, researchers implement it in the form of a network-based experiment for its evaluation, and then it needs to be validated through an available set of validation techniques, given below.

4.4.1. Simulation/emulation

It is a well-manageable and controllable framework that does network-based experiments on a single machine. These experiments are quite easy to configure and manage. In the detection of DDoS in SDN-enabled cloud, researchers have used various simulators, including NS3 [51], Mininet [50], CloudSimSDN [73] and a combination of two such as Weka [74] and Mininet. Table 8 shows the list of works that used these simulation and emulation tools to validate their work. In research articles, it has been found that Mininet alone is used by most researchers, i.e. more than 80%. The reason behind using this emulator may be its quickness for creating realistic virtual networks running actual kernel, switch and software application code on a personal computer. Also, it allows running unmodified code on virtual hardware on a simple PC. It offers convenience and realism at a very low cost. Also, many have used various attack and traffic generation tools to validate their work. The commonly used are Scapy [75], Hping3 [76], ID-ITG [36], Kali [77], Stacheldraht [66]. Moreover, authors have used datasets including CAIDA [78], DARPA 2009 [79], UNB-ISCX [80], CIC DoS 2019 [81], CICIDS2017 [82], Hibenx [83], ISCXIDS2012 [80], KDD Cup 99 [84] and CTU-13 [85] to evaluate their work.

4.4.2. Real testbed

This provides a real network, operating system, applications and platform which is best to carry out the experiments. All these experiments have some disadvantages also, such as adaptation of topology according to a new experiment is difficult and live experimentation of malicious software is quite hazardous as these programs can easily hide from experimentation setup. Many researchers have carried out their experiments using real environment including [37,44,64,66,67,92]. have used Cisco Nexus Switch 5672UP as an OpenFlow switch and Ryu Controller on Ryu-Processor: Intel core i7-7500U. They have used Edimax BR-6428NS Access point for network Connection and Quadcore Intel Xeon CPU for setting up the cloud environment. Apart from this, they have used six wireless VLANs that were set with various IoT sensors including Pi Matrix and piezoelectric sensor. On the other hand, [44] had set up a testbed for virtual networks that includes a router (HP MSR930-JG511 A), a physical switch (HP 3500 yl), an OpenStack platform (one controller node, one network node, an SDN controller (ONOS), and three compute nodes running OvS drivers on OpenFlow virtual switches for cloud networking).

Table 7
Detection rationale of various research articles.

Ref.	Detection scheme	Detection rationale
[35]	ST-GCN	Detection based on spatial and temporal features of the data
[38]	Fuzzy Hybrid with AI	The packets having lower number of successful packets, high packet loss, and high error rate are considered as malicious packets.
[39]	Moving Target Defense and Deception Technique	Creates uncertainty for the attackers by continuously and randomly modifying the attack surfaces.
[44]	Big data based — Apache Kafka, Apache Spark	Apache spark to find the optimum ML based technique from ML pipeline.
[54]	Shannon Entropy	Destination IP Address, Destination Port Address and Source Port Address
[60]	SVM, c4.5 Decision Tree and Naïve Bayes	Relative distribution of packet interval, number of packets, duration of the flow and relative distribution of match bytes
[63]	Support Vector Machine	If the output of proposed SVM function is -1 , then it is a DDoS attack packet, otherwise, it is normal packet.
[67]	SOM	The scheme preserves the topological structure of input data traffic flows using Euclidean Distance
[69]	Adaptive Fast Entropy	When the count of packets reaches the threshold
[70]	Entropy	Destination Address
[72]	Cumulative Sum (CuSum)	When the volume of packets flowing per unit get increased from the average of historical observations

Table 8
The description of Research Work incorporated simulation/emulation tools, traffic generation tools and Datasets.

S. No.	Simulator/emulator	References
1	NS3 [51]	[38]
2	Mininet [50]	[36,39,45,47–49,54,57,58,60,61,68,71,72,86,87]
3	CloudSimSDN [73]	[88]
4	Weka [74] and Mininet	[59]
S. No.	Traffic Generation Tools	References
1	Scapy [75]	[48,58,60,68–70,72]
2	Hping3 [76]	[35,36,39,42,57]
3	ID-ITG [89]	[36]
4	Kali [77]	[48,55]
5	Stacheldraht [90]	[66]
S. No.	Datasets	References
1	CAIDA [78]	[35,72]
2	Hibench [91]	[44]
3	UNB-ISCX [80]	[59,64]
4	CICIDS2017 [64]	[36]
5	CIC DoS 2019 [81]	[36,60]
6	DARPA 2009 [79]	[72]
7	ISCXIDS2012 [80]	[59]
8	KDDCup99 [84]	[45,65]
9	CTU-13 [85]	[59]

4.4.3. Mathematical model

These are theoretic in nature. In this, the proposed techniques have been presented symbolically and then validated mathematically. Chowdhary et al. [86] have used game theoretic-based validation of the proposed defense technique.

Discussion: Most research has validated their defense approaches using various simulators such as Mininet and NS3. Because of a virtual machine's limited capacity and resources, it cannot produce as accurate results as in a real environment. Only a few have used real setups, including hardware switches. Still, again, they have chosen a small topology which cannot predict the behavior of a real complex network under DDoS attack. So, research work in a real environment is a major research issue. Most researchers have used various traffic generation tools to generate attack traffic, including Scapy and Hping3. Although many have used datasets such as DARPA and KDD Cup99 for malicious

traffic, for normal traffic, again, they have used traffic generation tools. The traffic generation tools cannot generate the same traffic as real ones. So, evaluating defense techniques with a real datasets containing both attack and normal traffic is another research area.

4.5. Classification based on controller scalability

In count, the number of controllers may vary according to the scale and complexity of a network. Many researchers have demonstrated their work, and proposed defense techniques, either with the use of a single controller or multiple controllers.

4.5.1. Single controller architecture

The majority of the researchers have worked with a single controller. For this architecture, where POX [49,57,60,61,68,72], OpenDaylight [86] and NOX [66]. They have shown the validation of their work by making a single controller as a DDoS target or providing security to the network using an alone controller. However, the use of a single controller in a practical scenario is not feasible because it acts as a single point of failure [64].

4.5.2. Multiple controllers architecture

Only a few researchers worked with multiple controllers. [64] have used Ryu controllers, whereas [54] have used Floodlight and [87] have used ONOS controllers. Multiple Ryu controller architecture is opted by [64], at the local and global levels. A local SDN controller contains a detection and mitigation scheme, where the malicious IoT device is put it into the blacklist to block it. This information is forwarded to the controller at the global level and then that controller sends this information to the local controllers. Therefore, the multiple controller architecture is quite effective to detect DDoS attacks on a large scale.

Discussion: A single controller architecture is used by the majority of researchers as given in the previous section. A single controller in a network is not only a single point of failure but also suffers from security as POX and Ryu lacks Transport Layer Security (TLS). Although a few have used multi-controller architecture but lack synchronization and consistency. Therefore, defense using distributed controller architecture to provide a reliable network is a major research issue.

4.6. Classification based on defense deployment

The deployment of detection and mitigation technique is deployed at various spots in SDN-enabled cloud environment including controller, switch and cloud server. Majority of researchers have deployed their defense techniques on *SDN controller* [35,38,39,42,44,47,48,57,59–61,64,68,69,72,86,93]. The reason behind this is a controller has a global view of the whole network and locating and defending attackers from this spot is easy and efficient. Also, some have deployed their techniques in combination of *controller with cloud server* [36,55,66,92] and *switch* [49,54,63,70]. Apart from this, researchers are working with *Programming Protocol-independent Packet Processor (P4) switches* [94] to defend against DDoS attacks in SDN environment. P4 is a technology for network devices, especially for data plane devices to access, filter and evaluate packets. P4 is a domain-specific language that is developed to program the behavior of any switch whether it is operated at the local level or global level (using SDN controller).

Discussion: As shown, maximum researchers have deployed their defense techniques on SDN controllers only as the SDN environment is using dumb switches. This defense tactic is overwhelming the burden on the controller as well as the southbound interface. The resources and memory of the controller is been consumed at a high pace as it is responsible for the task of network supervision as well as detection of DDoS attacks. Moreover, a controller needs to accumulate the flow of statistical information from the switches which exhausts the bandwidth of the southbound interface. Apart from this, this practice introduces delay as first switches send the flow information to the controller and then only, the detection technique on the controller inspects the flow and makes decisions. However, some of the researchers have deployed their defense techniques on the cloud server and switch, but this increases the complexity and cost. So, adding intelligence to switches without affecting the cost and complexity of the network is the major research area.

5. General issues & challenges

5.1. Handling big data traffic for real-time detection of DDoS attack

SDN has various properties that can benefit big data applications, including data processing in cloud data centers, accurate data reachability, flexible assignment of tasks in Hadoop and scalable big data architectures [95]. As big data is usually processed in cloud data centers, SDN-enabled Clouds provide more advantages to big data management and transmission and preserve various big data applications' Service Level Agreements (SLAs). Despite all these benefits, DDoS attacks are still a major concern in big data technology. It is difficult for traditional defense mechanisms to detect normal and malicious traffic from large-scale network traffic [96].

In [97], the authors applied Multivariate Dimensionality Reduction Analysis (MDRA) and Mahalanobis distance (MD) to classify normal and attack traffic in Big data. Also, the advantage of Principal Component Analysis (PCA) to decrease the dimensionality of multiple features on KDD CUP 1999 dataset is used. Their detection mechanism is highly precise in True Negative Rate (TNR), with minimal processing time and memory resource consumption. Their technique is effective against DNS amplification, UDP flood, SYN flood and TCP flood DDoS attacks. Also, authors of [78] have shown concern about Hadoop's lack of acceptance due to serious security issues, including service availability. So, a simple concept of redundancy is carried out, where several data replicas have been stored on multiple data nodes under name nodes. The authors [98] have used the network flow feature sequence prediction technique to decrease the response time and increase the detection rate of DDoS attacks on CAIDA DDoS Attack 2007 dataset.

5.2. IoT-based DDoS attacks — a big issue

The IoT industry is expanding significantly where the devices related to our daily life and crucial specialized infrastructure systems receive unified connections. There is a huge expansion in the market of IoT devices, but these devices' security is still in the unfavorable stage. Therefore, attackers target these less secured IoT devices and compromise them to generate huge traffic towards a single point. Mirai Botnet is one example, which was composed of 65,000 devices and targets the French web host and cloud service providers with traffic of 1.1 Tbps in 2016 [99]. Not only do these devices act as a weapon in DDoS attacks, but also remain a potential target of attackers. The same Mirai Botnet targets the smart heating systems in Finland which leads to the continuous reboot of heating systems till the whole system came into crashes. The primary reason for the low security of these devices is their low computational power and limited storage which makes it difficult to implement practical solutions on them. Even if, lightweight solutions be imposed on them, still it is easy for attackers to bypass them. Therefore, it is a serious issue to protect these devices from DDoS attacks and detect IoT DDoS traffic in the network.

Authors [98] have proposed a complete DDoS defense mechanism using machine learning techniques including Long Short-term memory (LSTM) and convolutional neural network (CNN). They have deployed their technique on edge servers for early detection of DDoS attacks. they listed ten attack-relevant features such as Flow Bytes/s, Flow IAT Mean, Subflow Bwd Pkts and Fwd IAT Mean, and information gain is used to measure the entropy value of these features. The authors have used traffic generated by simulators and the CICDDoS2019 dataset to evaluate their results. Also, many authors have used SDN to detect IoT-based DDoS attacks and in [100], they have used entropy-based solutions deployed on the SDN data plane. They used the Mininet emulator and Ryu controller to conduct the experiments. Also, two datasets called BigFlows and Bot-IoT have been used to evaluate the proposed defense mechanism. However, they have found that their application gives inappropriate results when the window size is large which results to affect the responsiveness of switches according to the controller. Similarly, an application is built by [101] that operates on the application layer and mitigates the DDoS traffic with the insertion of policies to block access to the controller and switches. The technique is based on the threshold as it gets active when the count for Packet_in messages increases a predefined limit value. They have used mininet for their technique experimentation. Also, they used the iPerf synthetic traffic technique. The only problem is this technique is unreliable when the attacker spoofs the IP address.

5.3. Challenges in distributed SDN controller network

Distributed SDN Controllers are getting huge popularity in the research community. Even, Google has recently implemented a distributed SDN platform in their datacenters including Jupiter and wide area B4 networks [102]. Still, there are some challenges that need to be solved [103]. Firstly, accurate global network monitoring is an issue. Multiple controllers need to communicate their network updations to each other in order to have a consistent global view but, transferring the latest network information while maintaining good performance is a problem. If the controllers have dissimilar information about each other's domain, may lead to unexpected application operations.

Also, fault tolerance in distributed SDN control network is an important aspect. A simple master-slave approach used in a centralized SDN control network is not effective to make a reliable distributed SDN control network. A simple self-healing approach, where the control of the down controller is distributed in active controllers, has the burden of maintaining the network state and division of load in the remaining controllers.

Lastly, security related to authentication in distributed architecture is the researcher's concern. With a lack of proper authentication mechanisms, an attacker can launch its own node that acts as

an SDN controller that can violate the normal functioning of the whole network. Similarly, the communication channel between various controllers should be secure.

5.4. Handling flash events (FEs) in SDN enabled cloud

Flash event is when a huge number of normal users' access to web and network services results in performance degradation. Although DDoS and Flash Events are equivalent, the users in Flash Events are legitimate, and those in DDoS are attackers. Some authors have highlighted the discrimination of FEs and DDoS in the SDN scenario. General Entropy and Generalized Information Distance are mentioned by [104] to discriminate DDoS and FEs in SDN. Similarly, an improved Shannon Entropy is carried out by [105], where they have utilized the flow table characteristics to extract, analyze and classify the traffic. Still, more efforts are required to distinguish DDoS and Flash Events in SDN.

5.5. Up-gradation of SDN in internet service providers (ISPs)

SDN provides a global view of the whole network and dynamic management of routing paths and eliminates the shortcomings of existing IP protocols, including OSPF (Open Shortest Path First) [106] and MPLS (Multiprotocol Label Switching) [107]. SDN facilitates advanced Traffic Engineering (TE) approaches that provide dynamic network adaptations and accurate routing decisions. Also, the central controller ensures the QoS objectives regarding delay and bandwidth, as the controller is responsible for instant network configuration changes [108].

However, the upgradation to SDN requires huge efforts. Firstly, upgrading the large and expensive core networks of ISPs in a single go is impossible because of the operational, performance and security risks involved. Secondly, it is an expensive process, and upgradation requires core network components comprised of high cost. So, a hybrid network containing SDN and Non-SDN nodes is the leftover option where some part of the network adapts SDN while the remaining operates similarly. Still, some unanswered questions include how many nodes are required to upgrade and how to select them. and what should the time gap be in these nodes' upgradation? [109].

5.6. Mitigation in SDN — A challenge

The centralized controller of the SDN network has a complete view of the network, and it is easy to locate the malicious host or switch compared to a traditional network [110]. Most researchers insert new flow rules in switches to migrate and discard malicious traffic [100]. Many strategies include dropping traffic [101], blocking ports and migrating traffic [48], maintaining a list of blocked IPs [42], using a Game Theoretical Approach [111], and Fog Computing [43]. Although all the techniques, including blocking ports and discarding packets, are highly effective for immediately reducing attack impact, these lead to huge collateral damage in the network. Blocking any host or switch under attack may also block legitimate traffic.

Moreover, false attack detection prevents normal traffic from accessing network services. Also, the redirection method [112] diverts the legitimate traffic from one server under a DDoS attack to another so that the normal clients will not suffer. This approach increases the overall complexity of the whole network and leads to more processing time. All these techniques are relatively less impactful in defending against DDoS attacks in today's data centers because various data center applications have diversity in traffic volume and interruption tolerance. The above-mentioned techniques provide more flexibility in mitigating DDoS attacks than non-SDN networks. At the same time, these are ineffective in terms of detection of application-specific detection.

5.7. Lack of SDN-based benchmark datasets

A dataset provides a controlled environment to examine a proposed defense approach and to predict its results in real scenarios. One of the major issues is that there is no standard dataset for validating DDoS Defense systems, particularly in SDN. The traditional datasets such as KDD-99 [84] and CAIDA [78] have become obsolete and are inappropriate for drastically changing networks, including SDN. The primary reason is the limited availability of detailed public information regarding various attack scenarios. Many organizations and businesses are reluctant to reveal attack-related information in society for their public reputation. Therefore, generating a comprehensive dataset and effective defense system with partial information about attacks is difficult.

Also, testing these defense approaches is a concern to accurately validate defense techniques. Although there are simulators that have been used by researchers, including Mininet [50], EstiNet [113] and CloudSim SDN [114], these all work on a small scale. A real environment with the ability to live distributed experiments is required to check the working and performance of defense techniques.

6. Conclusion

SDN's concept of separating the control plane from the data plane greatly benefits various technologies, including cloud computing, IoT and fog computing. The SDN-enabled cloud is providing a solution to the mishandling of a plethora of data and traffic among the network of DCNs. The biggest concern is the security of the SDN-enabled cloud, where DDoS attack is becoming a hurdle for the continuous availability of cloud service providers.

In this review article, the attack and defence taxonomies of SDN-enabled cloud have been proposed. A taxonomy of DDoS attacks in the SDN-enabled cloud is showing various DDoS attacks, such as HTTP flood, TCP-SYN flood and UDP flood, etc., according to the layers of SDN-enabled cloud. It is observed that a maximum of researchers have highlighted the impact of DDoS attacks on controllers only, and only a few have emphasized various communication links (Northbound API, Southbound API and East-West bound API). Therefore, there is a huge requirement to protect these crucial links for the smooth functioning of networks. Also, a detailed taxonomy of DDoS defense in SDN-enabled cloud mentions six dimensions of defense, namely defense in collaboration with modern technologies, based on defense activities, based on the validation process, based on defense technology, based on controller scalability and based on defense deployment. For defense technology, it has been analyzed that a maximum of work has been done using machine learning, whereas SVM, particularly, has been used by many researchers. Also, security has been given to the controller and control layer only; minimal work has been done to protect the infrastructure and application layers, including different communication links. Moreover, detecting and mitigating slow-rate DDoS attacks is the major research issue. Most of the related work relies on simulators/emulators, attack traffic generation tools and synthetic Datasets for validation. An evaluation using real-bed topologies and real datasets is quite rare. The research gaps and discussions shown in the article will help the research fellows to understand the state-of-the-art. The challenges include the burden on the controller for attack defense, the unreliability of threshold-based defence schemes, inefficient results using traffic generation tools, dependency on simulators and emulators, using a single controller network, etc. Moreover, the general issues and challenges related to SDN-enabled cloud include handling big data traffic, IoT-based DDoS attacks, challenges on distributed SDN controllers, handling flash events, challenges of mitigation in SDN, etc. These research gaps and general issues remain a hurdle for providing security against DDoS attacks in the SDN-enabled cloud.

7. Limitation and future work

In this article, we have formulated the taxonomies of DDoS attack and defense mechanisms in SDN-enabled cloud. The taxonomies formulated presents only the theoretical perspectives of DDoS attacks and their defense mechanisms based on their *modus operandi*. We are highly motivated to develop an effective, efficient and robust DDoS detection and mitigation framework for real-world SDN enabled cloud testbed experimentations. Moreover, more details regarding the SDN related benchmark datasets can be explored for understanding the DDoS attack in SDN-Enabled Cloud. Emergence and Integration of technologies like Edge computing, Intent based networking, multi cloud environment and Blockchain can significantly change the study of taxonomical aspects of DDoS attacks and their defense mechanism in SDN-Enabled cloud.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] T. Cisco, A. Internet, Cisco: 2020 CISO benchmark report, *Comput. Fraud Secur.* 2020 (2020) 4, [http://dx.doi.org/10.1016/s1361-3723\(20\)30026-9](http://dx.doi.org/10.1016/s1361-3723(20)30026-9).
- [2] Number of internet and social media users worldwide as of July 2023, 2023, <https://www.statista.com/statistics/617136/digital-population-worldwide/>. (Accessed: 23 January 2023).
- [3] Telecom Regulatory Authority of India, The Indian telecom services performance indicators, 2023, pp. 1–159, http://www.trai.gov.in/WriteReadData/PIRReport/Documents/Indicator_Reports-Dec-14=08052015.pdf. (Accessed: 23 January 2023).
- [4] S. Kemp, DIGITAL 2020: July global statshot, 2023, <https://datareportal.com/reports/digital-2020-july-global-statshot>. (Accessed: 23 January 2023).
- [5] FLEXERA, State of the cloud report: Devops trends, 2021, pp. 1–19.
- [6] J. Singh, S. Behal, Detection and mitigation of DDoS attacks in SDN: A comprehensive review, research challenges and future directions, *Comp. Sci. Rev.* 37 (2020) 100279, <http://dx.doi.org/10.1016/j.cosrev.2020.100279>.
- [7] J. Son, R. Buyya, A taxonomy of Software-Defined Networking (SDN)-enabled cloud computing, *ACM Comput. Surv.* 51 (2019) 1–36, <http://dx.doi.org/10.1145/3190617>.
- [8] M.S. Bonfim, K.L. Dias, S.F.L. Fernandes, Integrated NFV/SDN architectures, *ACM Comput. Surv.* 51 (2019) 1–39, <http://dx.doi.org/10.1145/3172866>.
- [9] P.P. Ray, N. Kumar, SDN/NFV architectures for edge-cloud oriented IoT: A systematic review, *Comput. Commun.* 169 (2021) 129–153, <http://dx.doi.org/10.1016/j.comcom.2021.01.018>.
- [10] N. Anerousis, P. Chemouil, A.A. Lazar, N. Mihai, S.B. Weinstein, The origin and evolution of open programmable networks and SDN, *IEEE Commun. Surv. Tutor.* 23 (2021) 1956–1971, <http://dx.doi.org/10.1109/COMST.2021.3060582>.
- [11] M.A.B. Pérez, N.Y.S. Losada, E.R.S. nchez, G.M. Gaona, State of the art in Software Defined Networking (SDN), *Visión electrónica* 13 (2019) 178–194, <http://dx.doi.org/10.14483/22484728.14424>.
- [12] A.L. Aliyu, A. Aneiba, M. Patwary, P. Bull, A trust management framework for Software Defined Network (SDN) controller and network applications, *Comput. Netw.* 181 (2020) 107421, <http://dx.doi.org/10.1016/j.comnet.2020.107421>.
- [13] A.F. Abdullah, F.M. Salem, A. Tammam, M.H.A. Azeem, Performance analysis and evaluation of Software Defined Networking controllers against denial of service attacks, *J. Phys. Conf. Ser.* 1447 (1) (2020) 012007, <http://dx.doi.org/10.1088/1742-6596/1447/1/012007>.
- [14] T. Hu, Z. Zhang, P. Yi, D. Liang, Z. Li, Q. Ren, Y. Hu, J. Lan, SEAPP: A secure application management framework based on REST API access control in SDN-enabled cloud environment, *J. Parallel Distrib. Comput.* 147 (2021) 108–123, <http://dx.doi.org/10.1016/j.jpdc.2020.09.006>.
- [15] M. Brooks, B. Yang, A man-in-the-middle attack against OpenDayLight SDN controller, in: *Proceedings of the 4th Annual ACM Conference on Research in Information Technology*, 2015, pp. 45–49, <http://dx.doi.org/10.1145/2808062.2808073>.
- [16] Marketsandmarkets, Software-defined networking market, 2023, <https://www.marketsandmarkets.com/Market-Reports/software-defined-networking-sdn-market-655.html>. (Accessed: 06 October 2023).
- [17] GlobalMarketinsights, Software-defined networking market, 2023, <https://www.gminsights.com/industry-analysis/software-defined-networking-sdn-market>. (Accessed: 06 October 2023).
- [18] S. Maurya, S. Kumar, U. Garg, M. Kumar, An efficient framework for detection and classification of IoT botnet traffic, *ECS Sensors Plus* 1 (2022) 026401, <http://dx.doi.org/10.1149/2754-2726/ac7abc>.
- [19] J. David, C. Thomas, Discriminating flash crowds from DDoS attacks using efficient thresholding algorithm, *J. Parallel Distrib. Comput.* 152 (2021) 79–87, <http://dx.doi.org/10.1016/j.jpdc.2021.02.019>.
- [20] A. Singhal, Findings from 2nd half 2022 unveiling the new threat landscape - DDoS threat intelligence report, 2022, URL <https://www.netscout.com/threatreport/wp-content/uploads/2023/04/Threat-Report-2H2022.pdf>.
- [21] R. Deb, S. Roy, A comprehensive survey of vulnerability and information security in SDN, *Comput. Netw.* 206 (2022) 108802, <http://dx.doi.org/10.1016/j.comnet.2022.108802>.
- [22] S. Karnani, H.K. Shakya, Mitigation strategies for distributed denial of service (DDoS) in SDN: A survey and taxonomy, *Inform. Secur. J.: Global Perspect.* 32 (2023) 444–468, <http://dx.doi.org/10.1080/19393555.2022.2111004>.
- [23] T.E. Ali, Y.W. Chong, S. Manickam, Machine learning techniques to detect a DDoS attack in SDN: A systematic review, *Appl. Sci. (Switzerland)* 13 (2023) <http://dx.doi.org/10.3390/app13053183>.
- [24] A.A. Alashhab, M.S.M. Zahid, M.A. Azim, M.Y. Daha, B. Isyaku, S. Ali, A survey of low rate DDoS detection techniques based on machine learning in software-defined networks, *Symmetry* 14 (2022) 1563, <http://dx.doi.org/10.3390/sym14081563>.
- [25] J.F. Balarezo, S. Wang, K.G. Chavez, A. Al-Hourani, S. Kandeepan, A survey on DoS/DDoS attacks mathematical modelling for traditional, SDN and virtual networks, *Eng. Sci. Technol. Int. J.* 31 (2022) 101065, <http://dx.doi.org/10.1016/j.jestech.2021.09.011>.
- [26] N. Ahmed, K.A. Bakar, F.T. Zuhra, T. Kehkashan, M.A. Mujahid, M.S. Rathore, M. Dawood, B. Isyaku, Security & privacy in Software Defined Networks, issues, challenges and cost of developed solutions: A systematic literature review, *Int. J. Wirel. Inf. Netw.* 29 (2022) 314–340, <http://dx.doi.org/10.1007/s10776-022-00561-y>.
- [27] Y. Maleh, Y. Qasmaoui, K.E. Gholami, A comprehensive survey on SDN security: threats, mitigations, and future directions, *J. Reliab. Intell. Environ.* (2022) <http://dx.doi.org/10.1007/s40860-022-00171-8>.
- [28] S. Dong, K. Abbas, R. Jain, A survey on distributed denial of service (DDoS) attacks in SDN and cloud computing environments, *IEEE Access* 7 (2019) 80813–80828, <http://dx.doi.org/10.1109/ACCESS.2019.2922196>.
- [29] A.A. Djergou, Y. Maleh, S. Mounir, Machine learning techniques for intrusion detection in SDN: A survey, in: *Lecture Notes in Networks and Systems*, 2022, pp. 460–473, http://dx.doi.org/10.1007/978-3-030-91738-8_42.
- [30] S. Kaur, K. Kumar, N. Aggarwal, G. Singh, A comprehensive survey of DDoS defense solutions in SDN: Taxonomy, research challenges, and future directions, *Comput. Secur.* 110 (2021) 102423, <http://dx.doi.org/10.1016/j.cose.2021.102423>.
- [31] OpenFlow. Open networking foundation (ONF), 2022, <https://www.opennetworking.org/>. (Accessed: 23 January 2022).
- [32] A.M. Abdelrahman, J.J. Rodrigues, M.M. Mahmoud, K. Saleem, A.K. Das, V. Korotaev, S.A. Kozlov, Software-defined networking security for private data center networks and clouds: Vulnerabilities, attacks, countermeasures, and solutions, *Int. J. Commun. Syst.* 34 (2021) <http://dx.doi.org/10.1002/dac.4706>.
- [33] M. Snehi, A. Bhandari, Vulnerability retrospection of security solutions for software-defined cyber – physical system against DDoS and IoT-DDoS attacks, *Comp. Sci. Rev.* 40 (2021) 100371, <http://dx.doi.org/10.1016/j.cosrev.2021.100371>.
- [34] A. Aljuhani, Machine learning approaches for combating distributed denial of service attacks in modern networking environments, *IEEE Access* 9 (2021) 42236–42264, <http://dx.doi.org/10.1109/ACCESS.2021.3062909>.
- [35] Y. Cao, H. Jiang, Y. Deng, J. Wu, P. Zhou, W. Luo, Detecting and mitigating DDoS attacks in SDN using spatial-temporal graph convolutional network, *IEEE Trans. Dependable Secure Comput.* 5971 (2021) 3855–3872, <http://dx.doi.org/10.1109/tdsc.2021.3108782>.
- [36] N.M. Yungaicela-Naula, C. Vargas-Rosales, J.A. Perez-Diaz, SDN-based architecture for transport and application layer DDoS attack detection by using machine and deep learning, *IEEE Access* 9 (2021) 108495–108512, <http://dx.doi.org/10.1109/access.2021.3101650>.
- [37] P. Harikrishna, A. Amuthan, Rival-model penalized self-organizing map enforced DDoS attack prevention mechanism for Software Defined Network-based cloud computing environment, *J. Parallel Distrib. Comput.* 154 (2021) 142–152, <http://dx.doi.org/10.1016/j.jpdc.2021.03.005>.
- [38] I.H. Abdulqader, D. Zou, I.T. Aziz, B. Yuan, W. Dai, Deployment of robust security scheme in SDN based 5G network over NFV enabled cloud environment, *IEEE Trans. Emerg. Top. Comput.* 9 (2021) 866–877, <http://dx.doi.org/10.1109/TETC.2018.2879714>.

- [39] Y. Zhou, G. Cheng, S. Yu, An SDN-enabled proactive defense framework for DDoS mitigation in IoT networks, *IEEE Trans. Inf. Forensics Secur.* 16 (2021) 5366–5380, <http://dx.doi.org/10.1109/TIFS.2021.3127009>.
- [40] P. Harikrishna, A. Amuthan, Rival model penalized self-organizing map enforced DDoS attack prevention mechanism for Software Defined Network-based cloud computing environment, *J. Parallel Distrib. Comput.* 154 (2021) 142–152, <http://dx.doi.org/10.1016/j.jpdc.2021.03.005>.
- [41] Openflow agent, 2011, https://www.arubanetworks.com/techdocs/ArubaOS_83_Web_Help/Content/ArubaFrameStyles/OF_Agent/ofa_overv.htm. (Accessed: 22 May 2021).
- [42] R. Priyadarshini, R.K. Barik, H. Dubey, Fog-SDN: A light mitigation scheme for DDoS attack in fog computing framework, *Int. J. Commun. Syst.* 33 (2020) 1–19, <http://dx.doi.org/10.1002/dac.4389>.
- [43] K.A. Sadiq, A. Thompson, O. Ayeni, Mitigating DDoS attacks in cloud network using fog and SDN: A conceptual security framework, *Int. J. Appl. Inform. Syst. (IJAIS)* 12 (2020) 11–16, <http://dx.doi.org/10.5120/ijais2020451877>.
- [44] P.T. Dinh, M. Park, BDF-SDN: A big data framework for DDoS attack detection in large-scale SDN-based cloud, in: 2021 IEEE Conference on Dependable and Secure Computing, DSC 2021, IEEE, Aizuwakamatsu, Fukushima, Japan, 2021, <http://dx.doi.org/10.1109/DSC49826.2021.9346269>.
- [45] M. Revathi, V.V. Ramalingam, B. Amutha, A machine learning based detection and mitigation of the DDOS attack by using SDN controller framework, *Wirel. Pers. Commun.* (2021) <http://dx.doi.org/10.1007/s11277-021-09071-1>.
- [46] A. Rahman, M.J. Islam, M.S.I. Khan, S. Kabir, A.I. Pritom, M.R. Karim, Block-SDN: Enhancing security of cloud storage through blockchain-based SDN in IoT network, in: 2020 2nd International Conference on Sustainable Technologies for Industry 4.0, STI 2020, 2020, <http://dx.doi.org/10.1109/STI50764.2020.9350419>.
- [47] K.M.S. Azad, N. Hossain, M.J. Islam, A. Rahman, S. Kabir, Preventive determination and avoidance of DDoS attack with sdn over the iot networks, in: 2021 International Conference on Automation, Control and Mechatronics for Industry 4.0, ACMI, IEEE, 2021, pp. 1–6, <http://dx.doi.org/10.13140/RG.2.2.12057.54886>.
- [48] A. Mishra, N. Gupta, B.B. Gupta, Defense mechanisms against DDoS attack based on entropy in SDN-cloud using POX controller, *Telecommun. Syst.* 77 (2021) 47–62, <http://dx.doi.org/10.1007/s11235-020-00747-w>.
- [49] K. Bhushan, B.B. Gupta, Distributed denial of service (DDoS) attack mitigation in software defined network (SDN)-based cloud computing environment, *J. Ambient Intell. Humaniz. Comput.* 10 (2019) 1985–1997, <http://dx.doi.org/10.1007/s12652-018-0800-9>.
- [50] MININET, 2021, <http://mininet.org/overview/>. (Accessed: 24 March 2021).
- [51] NS3- network simulator, 2023, <https://www.nsnam.org/>. (Accessed: 23 January 2023).
- [52] Floodlight, 2021, <http://www.projectfloodlight.org/>. (Accessed: 24 March 2021).
- [53] R. Chaganti, B. Bhushan, V. Ravi, The role of blockchain in DDoS attacks mitigation: Techniques, open challenges and future directions, 2022, URL <http://arxiv.org/abs/2202.03617>.
- [54] Z.A.E. Houda, A.S. Hafid, L. Khoukhi, Cochain-SC: An intra-and inter-domain DDoS mitigation scheme based on blockchain using SDN and smart contract, *IEEE Access* 7 (2019) 98893–98907, <http://dx.doi.org/10.1109/ACCESS.2019.2930715>.
- [55] Y. Xu, Y. Yu, H. Hong, Z. Sun, DDoS detection using a cloud-edge collaboration method based on entropy-measuring SOM and KD-tree in SDN, *Secur. Commun. Netw.* 2021 (2021) 1–16, <http://dx.doi.org/10.1155/2021/5594468>.
- [56] A. Saritha, B.R. Reddy, A.S. Babu, QEMDD: Quantum inspired ensemble model to detect and mitigate DDoS attacks at various layers of SDN architecture, *Wirel. Pers. Commun.* 127 (2021) 2365–2390, <http://dx.doi.org/10.1007/s11277-021-08805-5>.
- [57] S. Batool, F.Z. Khan, S.Q.A. Shah, M. Ahmed, R. Alroobaea, A.M. Baqasah, I. Ali, M.A. Raza, Lightweight statistical approach towards TCP SYN flood DDoS attack detection and mitigation in SDN environment, *Secur. Commun. Netw.* 2022 (2022) 1–14, <http://dx.doi.org/10.1155/2022/2593672>.
- [58] R.B. Shohani, S. Mostafavi, V. Hakami, A statistical model for early detection of DDoS attacks on random targets in SDN, *Wirel. Pers. Commun.* 120 (2021) 379–400, <http://dx.doi.org/10.1007/s11277-021-08465-5>.
- [59] A.B. Dehkordi, M.R. Soltanaghahi, F.Z. Boroujeni, The DDoS attacks detection through machine learning and statistical methods in SDN, *J. Supercomput.* 77 (2021) 2383–2415, <http://dx.doi.org/10.1007/s11227-020-03323-w>.
- [60] K.M. Sudar, P. Deepalakshmi, Flow-based detection and mitigation of low-rate DDOS attack in SDN environment using machine learning techniques, in: IoT and Analytics for Sensor Networks, in: Lecture Notes in Networks and Systems, 2022, pp. 193–205, http://dx.doi.org/10.1007/978-981-16-2919-8_18.
- [61] S.W. Tufa, M. Mengstie, H. Gebregziabher, B.R. Babu, Detecting DDoS attack using adaptive boosting with software defined network in cloud computing environment, *Revista Geintec-Gestao Inovacao E Tecnologias* 11 (2021) 3485–3494, <http://dx.doi.org/10.1109/SPIN.2018.8474062>.
- [62] Y. Xiao, Z. Jie Fan, A. Nayak, C. Xiang Tan, Discovery method for distributed denial-of-service attack behavior in SDNs using a feature-pattern graph model, *Front. Inf. Technol. Electron. Eng.* 20 (2019) 1195–1208, <http://dx.doi.org/10.1631/FITEE.1800436>.
- [63] L. Yang, H. Zhao, DDoS attack identification and defense using SDN based on machine learning method, in: Proceedings - 2018 15th International Symposium on Pervasive Systems, Algorithms and Networks, I-SPAN 2018, IEEE, 2019, pp. 174–178, <http://dx.doi.org/10.1109/I-SPAN.2018.00036>.
- [64] N. Ravi, S.M. Shalinie, Learning-driven detection and mitigation of DDoS attack in IoT via SDN-cloud architecture, *IEEE Internet Things J.* 7 (2020) 3559–3570, <http://dx.doi.org/10.1109/JIOT.2020.2973176>.
- [65] Y. Yang, K. Zheng, C. Wu, X. Niu, Y. Yang, Building an effective intrusion detection system using the modified density peak clustering algorithm and deep belief networks, *Appl. Sci.* 9 (2019) 238, <http://dx.doi.org/10.3390/app9020238>.
- [66] P. Harikrishna, A. Amuthan, SDN-based DDoS attack mitigation scheme using convolution recursively enhanced self organizing maps, *Sadhana - Acad. Proc. Eng. Sci.* 45 (2020) <http://dx.doi.org/10.1007/s12046-020-01353-x>.
- [67] H. Pillutla, A. Arjunan, Fuzzy self organizing maps-based DDoS mitigation mechanism for software defined networking in cloud computing, *J. Ambient Intell. Humaniz. Comput.* (2018) 1–13, <http://dx.doi.org/10.1007/s12652-018-0754-y>.
- [68] T.U. Nadim, Foysal, Towards autonomic entropy based approach for DDoS attack detection and mitigation using Software Defined Networking, in: 2021 International Conference on Automation, Control and Mechatronics for Industry 4.0, ACMI 2021, IEEE, 2021, pp. 8–9, <http://dx.doi.org/10.1109/ACMI53878.2021.9528288>.
- [69] H. Guesmi, L.A. Saïdane, Using SDN approach to secure cloud servers against flooding based DDoS attacks, in: Proceedings - 25th International Conference on Systems Engineering, ICSEng 2017, vol. 2017-Janua, 2017, pp. 309–315, <http://dx.doi.org/10.1109/ICSEng.2017.31>.
- [70] S.C. Tsai, I.H. Liu, C.T. Lu, C.H. Chang, J.S. Li, Defending cloud computing environment against the challenge of DDoS attacks based on Software Defined Network, *Smart Innov. Syst. Technol.* 63 (2017) 285–292, http://dx.doi.org/10.1007/978-3-319-50209-0_35.
- [71] N. Agrawal, S. Tapaswi, An SDN-assisted defense mechanism for the shrew DDoS attack in a cloud computing environment, *J. Netw. Syst. Manage.* 29 (2021) 1–28, <http://dx.doi.org/10.1007/s10922-020-09580-7>.
- [72] M. Conti, A. Gangwal, M.S. Gaur, A comprehensive and effective mechanism for DDoS detection in SDN, in: International Conference on Wireless and Mobile Computing, Networking and Communications, 2017, <http://dx.doi.org/10.1109/WIMOB.2017.8115796>.
- [73] CloudSimSDN, 2023, <https://github.com/Cloudslab/cloudsimsdn/releases>. (Accessed: 23 January 2023).
- [74] Weka, 2023, <https://www.weka.io/>. (Accessed: 23 January 2023).
- [75] Scapy, 2021, <https://scapy.net/download/>. (Accessed: 24 March 2021).
- [76] Hping, 2021, <http://www.hping.org/>. (Accessed: 24 March 2021).
- [77] M.H. Bhuyan, H.J. Kashyap, D.K. Bhattacharyya, J. Kalita, Detecting distributed denial of service attacks: Methods, tools and future directions, *Comput. J.* 57 (2012) 537–556, <http://dx.doi.org/10.1093/comjnl/bxt031>.
- [78] The CAIDA “DDoS attack 2007” dataset, 2023, https://www.caida.org/catalog/datasets/ddos-20070804_dataset. (Accessed: 23 January 2023).
- [79] M. Gharaibeh, C. Papadopoulos, DARPA-2009 intrusion detection dataset report, 2023, <https://ant.isi.edu/~johnh/PAPERS/Gharaibeh14a.pdf>. (Accessed: 23 January 2023).
- [80] Intrusion detection evaluation dataset (ISCIDX2012), 2023, Canadian Institute for Cybersecurity, <https://www.unb.ca/cic/datasets/ids.html>. (Accessed: 23 January 2023).
- [81] I. Sharafaldin, A.H. Lashkari, S. Hakak, A.A. Ghorbani, Developing realistic Distributed Denial of Service (DDoS) attack dataset and taxonomy, in: 2019 International Carnahan Conference on Security Technology, ICCST, IEEE, 2019, pp. 1–8, <http://dx.doi.org/10.1109/CCST.2019.8888419>.
- [82] I. Sharafaldin, A.H. Lashkari, A.A. Ghorbani, Toward generating a new intrusion detection dataset and intrusion traffic characterization, in: Proceedings of the 4th International Conference on Information Systems Security and Privacy, 2018, pp. 108–116, <http://dx.doi.org/10.5220/0006639801080116>.
- [83] S. Huang, J. Huang, J. Dai, T. Xie, B. Huang, The HiBench benchmark suite: Characterization of the MapReduce-based data analysis, in: 2010 IEEE 26th International Conference on Data Engineering Workshops, ICDEW 2010, IEEE, 2010, pp. 41–51, <http://dx.doi.org/10.1109/ICDEW.2010.5452747>.
- [84] KDD cup 1999 data, 2023, <http://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html>. (Accessed: 23 January 2023).
- [85] S. Garcia, M. Grill, J. Stiborek, A. Zunino, An empirical comparison of botnet detection methods, *Comput. Secur.* 45 (2014) 100–123, <http://dx.doi.org/10.1016/j.cose.2014.05.011>.
- [86] A. Chowdhary, S. Pisharody, A. Alshamrani, D. Huang, Dynamic game based security framework in SDN-enabled cloud networking environments, in: Proceedings of the ACM International Workshop on Security in Software Defined Networks and Network Function Virtualization, SDN-NFVSec '17, 2017, pp. 53–58, <http://dx.doi.org/10.1145/3040992.3040998>.
- [87] M.F. Hyder, T. Fatima, Towards crossfire distributed denial of service attack protection using intent-based moving target defense over software-defined networking, *IEEE Access* 9 (2021) 112792–112804, <http://dx.doi.org/10.1109/ACCESS.2021.3103845>.

- [88] G. Kaur, P. Gupta, Proposed optimization technique to detect DDOS attacks on software defined networks, *Int. J. Inform. Syst. Manag. Sci.* 2 (2) (2019).
- [89] G. Aceto, A. Botta, W. de Donato, A. Pescap, D-ITG: Distributed internet traffic generator, PIK - Praxis der Informationsverarbeitung und Kommunikation 36 (2013) <http://dx.doi.org/10.1515/pik-2012-0064>.
- [90] P.J. Criscuolo, Distributed denial of service tools trino, tribe flood network, tribe flood network 2000, and stacheldraht, 2000.
- [91] X. Huang, X. Du, B. Song, An effective DDoS defense scheme for SDN, in: IEEE International Conference on Communications, IEEE, Paris, France, 2017, <http://dx.doi.org/10.1109/ICC.2017.7997187>.
- [92] B. Wang, Y. Zheng, W. Lou, Y.T. Hou, DDoS attack protection in the era of cloud computing and software-defined networking, *Comput. Netw.* 81 (2015) 308–319, <http://dx.doi.org/10.1016/j.comnet.2015.02.026>.
- [93] S. Faizullah, M.A. Khan, A. Alzahrani, I. Khan, Permissioned blockchain-based security for SDN in IoT cloud networks, in: 2019 International Conference on Advances in the Emerging Computing Technologies, AECT 2019, 2020, <http://dx.doi.org/10.1109/AECT47998.2020.9194181>.
- [94] P4-open networking foundation, 2021, <http://opennetworking.org/p4/>. (Accessed: 24 March 2021).
- [95] M.J. Awan, U. Farooq, H.M.A. Babar, A. Yasin, H. Nobanee, M. Hussain, O. Hakeem, A.M. Zain, Real-time DDoS attack detection system using big data approach, *Sustainability* 13 (19) (2021) 10743, <http://dx.doi.org/10.3390/su131910743>.
- [96] S. Janakiraman, An integrated principal component and reduced multivariate data analysis technique for detecting DDoS attacks in big data federated clouds, *Int. J. Cloud Comput.* 10 (4) (2021) 339–355, <http://dx.doi.org/10.1504/IJCC.2021.119190>.
- [97] B. Jia, Y. Ma, X. Huang, Z. Lin, Y. Sun, A novel real-time DDoS attack detection mechanism based on MDRA algorithm in big data, *Math. Probl. Eng.* 2016 (2016) 1–10.
- [98] Y. Jia, F. Zhong, A. Alrawais, B. Gong, X. Cheng, FlowGuard: An intelligent edge defense mechanism against IoT DDoS attacks, *IEEE Internet Things J.* 7 (2020) 9552–9562, <http://dx.doi.org/10.1109/JIOT.2020.2993782>.
- [99] T.G. Palla, S. Tayeb, Intelligent Mirai malware detection for IoT nodes, *Electronics* 10 (11) (2021) 1241, <http://dx.doi.org/10.3390/electronics10111241>.
- [100] J. Galeano-Brajones, J. Carmona-Murillo, J.F. Valenzuela-Valdés, F. Luna-Valero, Detection and mitigation of DoS and DDoS attacks in iot-based stateful SDN: An experimental approach, *Sensors (Switzerland)* 20 (2020) 1–18, <http://dx.doi.org/10.3390/s20030816>.
- [101] S. Wang, K. Gomez, K. Sithamparanathan, M.R. Asghar, G. Russello, P. Zanna, Mitigating DDoS attacks in sdn-based iot networks leveraging secure control and data plane algorithm, *Appl. Sci. (Switzerland)* 11 (2021) 1–27, <http://dx.doi.org/10.3390/app11030929>.
- [102] A.D. Ferguson, S. Gribble, Orion: Google's software-defined networking control plane proceedings of the 18th USENIX symposium on orion: Google's software-defined networking control plane, in: Proceedings of NSDI 2021: 18th USENIX Symposium on Networked Systems Design and Implementation, 2021, pp. 83–98.
- [103] S. Ahmad, A.H. Mir, Scalability, consistency, reliability and security in SDN controllers: A survey of diverse SDN controllers, *J. Netw. Syst. Manage.* 29 (2021) 1–59, <http://dx.doi.org/10.1007/s10922-020-09575-4>.
- [104] K.S. Sahoo, M. Tiwary, B. Sahoo, Detection of high rate DDoS attack from flash events using information metrics in Software Defined Networks, in: 2018 10th International Conference on Communication Systems and Networks, COMSNETS, IEEE, 2018, pp. 421–424, <http://dx.doi.org/10.1109/COMSNETS.2018.8328233>, URL <http://ieeexplore.ieee.org/document/8328233/>.
- [105] G. Sun, W. Jiang, Y. Gu, D. Ren, H. Li, DDoS attacks and flash event detection based on flow characteristics in SDN, in: Proceedings of AVSS 2018 - 2018 15th IEEE International Conference on Advanced Video and Signal-Based Surveillance, 2019, pp. 3–8, <http://dx.doi.org/10.1109/AVSS.2018.8639103>.
- [106] OSPF design guide, 2023, <https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/7039-1.html>. (Accessed: 23 January 2023).
- [107] What is MPLS - Multiprotocol label switching, 2023, <https://www.cisco.com/c/en/in/products/ios-nx-os-software/multiprotocol-label-switching-mpls/index.html>. (Accessed: 23 January 2023).
- [108] D. Kafetzis, S. Vassilaras, G. Vardoulas, I. Koutsopoulos, Software-defined networking meets software-defined radio in mobile ad hoc networks: State of the art and future directions, *IEEE Access* 10 (2022) 9989–10014, <http://dx.doi.org/10.1109/ACCESS.2022.3144072>.
- [109] K. Poularakis, G. Iosifidis, G. Smaragdakis, L. Tassioulas, Optimizing gradual SDN upgrades in ISP networks, *IEEE/ACM Trans. Netw.* 27 (2019) 288–301, <http://dx.doi.org/10.1109/TNET.2018.2890248>.
- [110] F.S. Dantas Silva, E. Silva, E.P. Neto, M. Lemos, A.J. Venancio Neto, F. Esposito, A taxonomy of DDoS attack mitigation approaches featured by SDN technologies in IoT scenarios, *Sensors* 20 (11) (2020) 3078, <http://dx.doi.org/10.3390/s20113078>.
- [111] M.V.D. Assis, M.P. Novaes, C.B. Zerbin, L.F. Carvalho, T. Abrao, M.L. Proenca, Fast defense system against attacks in Software Defined Networks, *IEEE Access* 6 (2018) 69620–69639, <http://dx.doi.org/10.1109/ACCESS.2018.2878576>.
- [112] M. Aslam, D. Ye, A. Tariq, M. Asad, M. Hanif, D. Ndzi, S.A. Chelloug, M.A. Elaziz, M.A. Al-Qaness, S.F. Jilani, Adaptive machine learning based distributed denial-of-services attacks detection and mitigation system for SDN-enabled IoT, *Sensors* 22 (7) (2022) 2697, <http://dx.doi.org/10.3390/s22072697>.
- [113] EstiNet, 2023, https://www.estinet.com/ns/?page_id=23355. (Accessed: 23 January 2023).
- [114] J. Son, A.V. Dastjerdi, R.N. Calheiros, X. Ji, Y. Yoon, R. Buyya, CloudSimSDN: Modeling and simulation of software-defined cloud data centers, in: Proceedings - 2015 IEEE/ACM 15th International Symposium on Cluster, Cloud, and Grid Computing, CCGrid 2015, 2015, pp. 475–484, <http://dx.doi.org/10.1109/CCGrid.2015.87>.